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Process Mining Tutorial

Adapted from
fluxicon.com/academic/material/

Edited by Andrea Vandin, SSSA

Goals of this tutorial

- Understand the phases of **process mining** analysis

- Be able to **get started and play** around with your own data

Outline

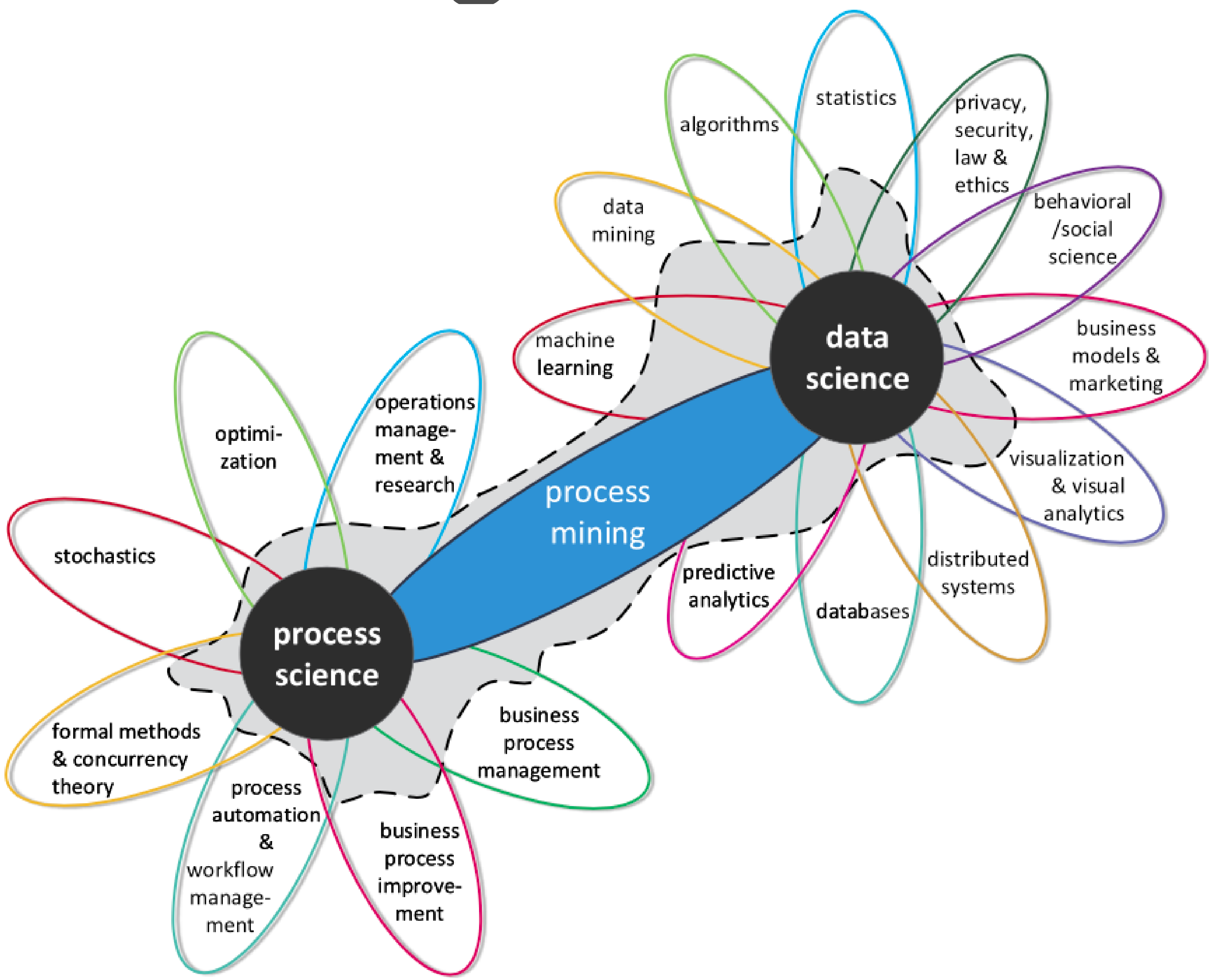
0. **What is Process mining?**
1. Example Scenario
2. Roadmap
3. Hands-on Session
4. Take-away Points

What is Process Mining?

Process mining is **a family of techniques linking data science and process management to support the analysis of operational processes based on event logs.**

The goal of process mining is to turn event data into insights and actions. Process mining is an integral part of data science, fueled by the availability of data and the desire to improve processes.

E.g., *process mining* uses event data to automatically *discover* a process model by observing events recorded by some enterprise system



Van Der Aalst, W., et al. (2011, August).
Process mining manifesto. In *International Conference on Business Process Management* (pp. 169-194).
Springer

What can we do with Process Mining?

With PM we can find the actual process is different from the one we designed!

What can we do with Process Mining?

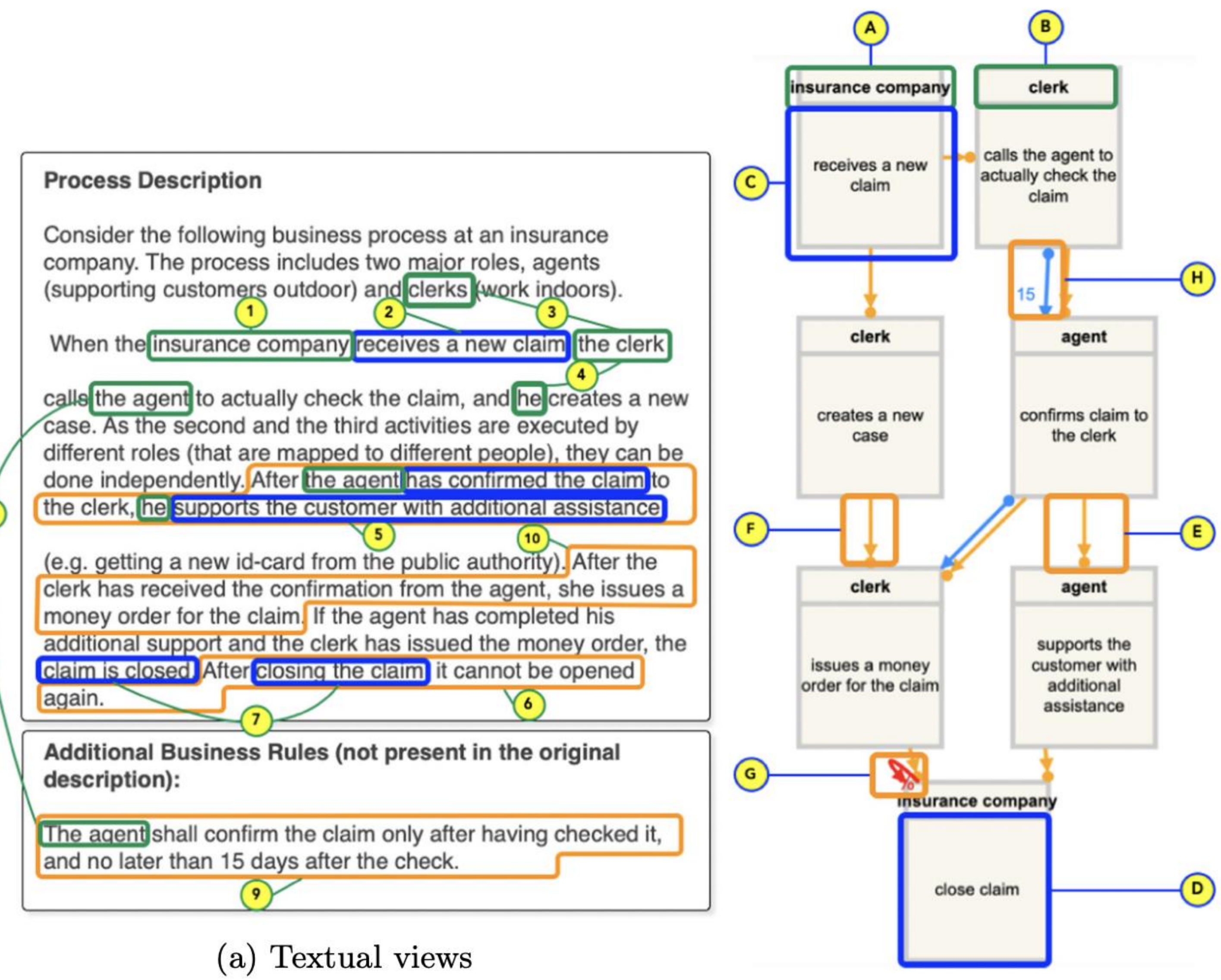
With PM we can find the actual process is different from the one we designed!

Our traces are the process logs…

We will see a simple example (purchasing process)

However, PM has been applied to several domains, e.g., the **legal** one

Declarative Process Discovery: Linking Process and Textual Views



(a) Textual views

(b) Model view as a DCR graph

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How efficient is a law court?

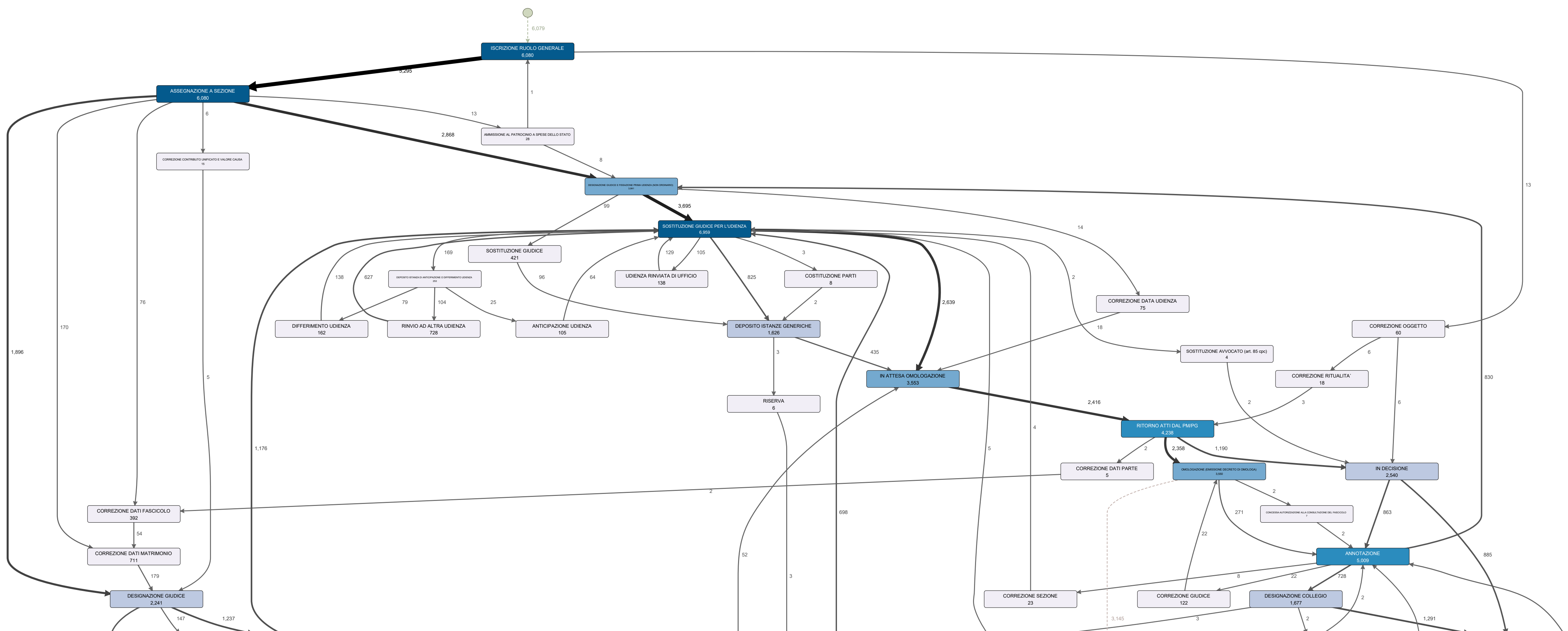
Process Mining for legal Courts: Visualising, analysing and comparing Italian divorce proceedings ☆

ELSEVIER

Computer Law & Security Review

Volume 59, November 2025, 106210

Vittoria Caponecchia, Bernardo D'Agostino, Sima Sarv Ahrabi, Giovanni Comandè, Daniele Licari, Andrea Vandin



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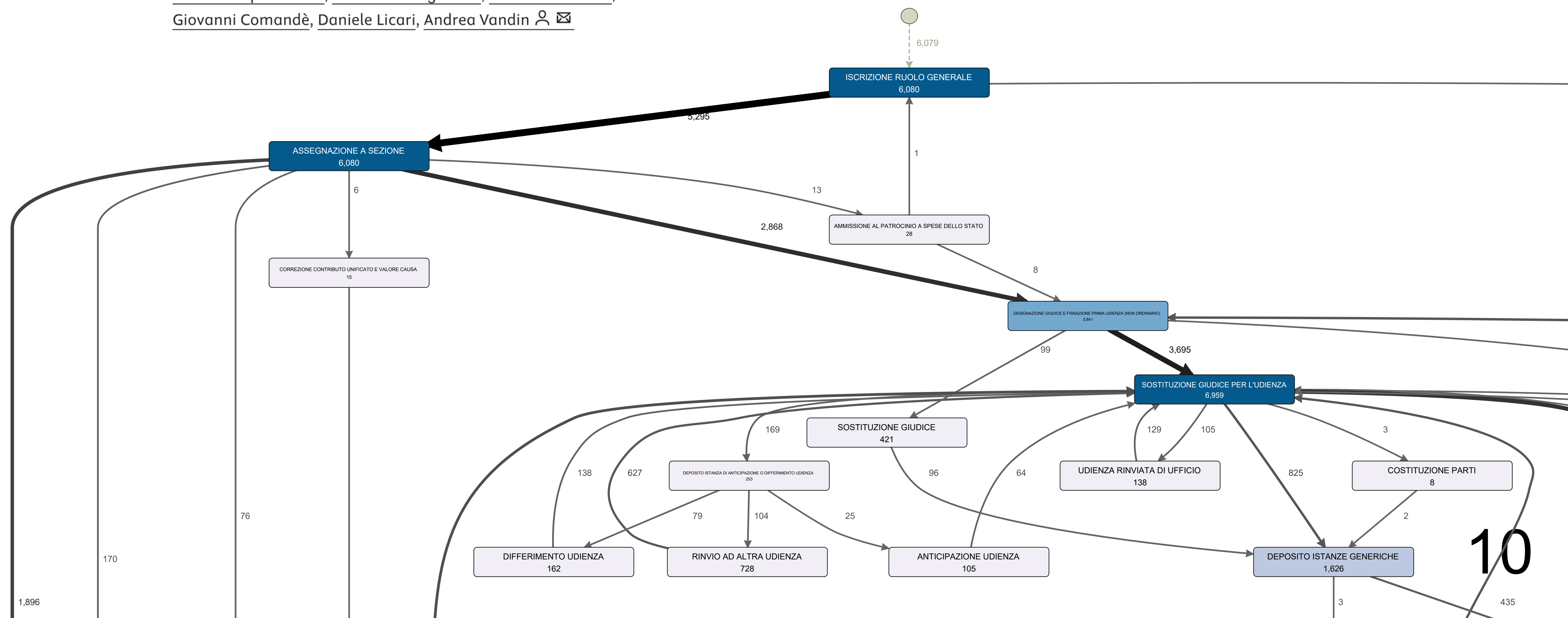


We will see a simple example (purchasing process)

However, PM has been applied to several domains, e.g., the **legal** one

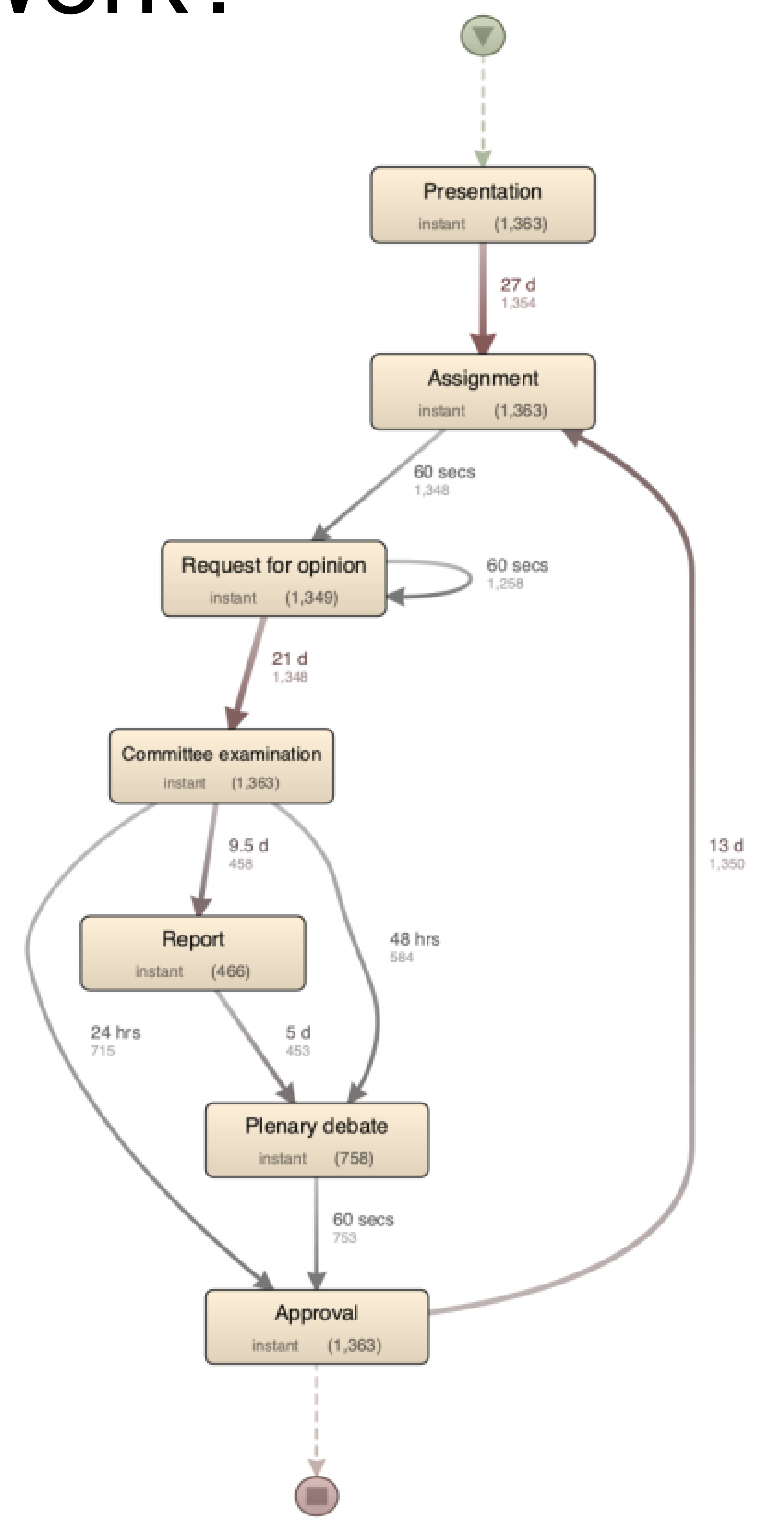
How does the Italian parliament work?

From Text to Process: Leveraging LLMs to Unveil Italian Lawmaking

Matilde Contestabile[1], Chiara Ferrara[1], Alberto Giovannetti[1], Giovanni Parrillo[1], and Andrea Vandin[1,2]

The ProLiFIC dataset: Leveraging LLMs to Unveil the Italian Lawmaking Process

Matilde Contestabile[1], Chiara Ferrara[1], Alberto Giovannetti[1], Giovanni Parrillo[1] and Andrea Vandin[1,2]



(a) Maps for X and XI legislatures

We will see a simple example (purchasing process)

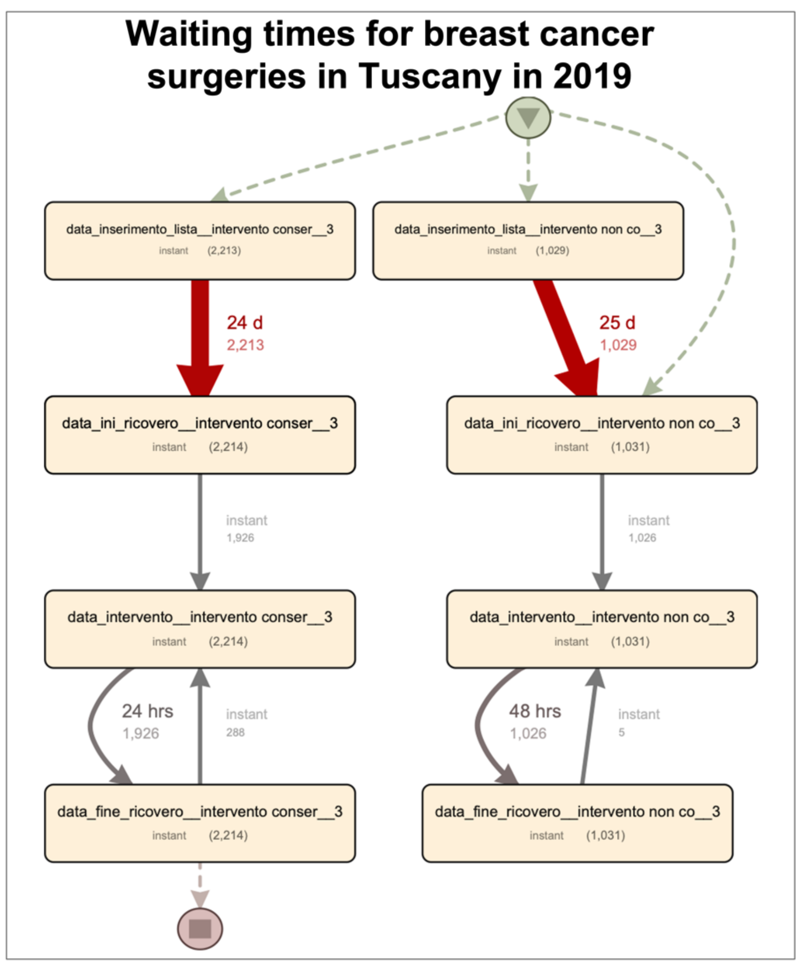
However, PM has been applied to several domains, e.g., the **healthcare** one

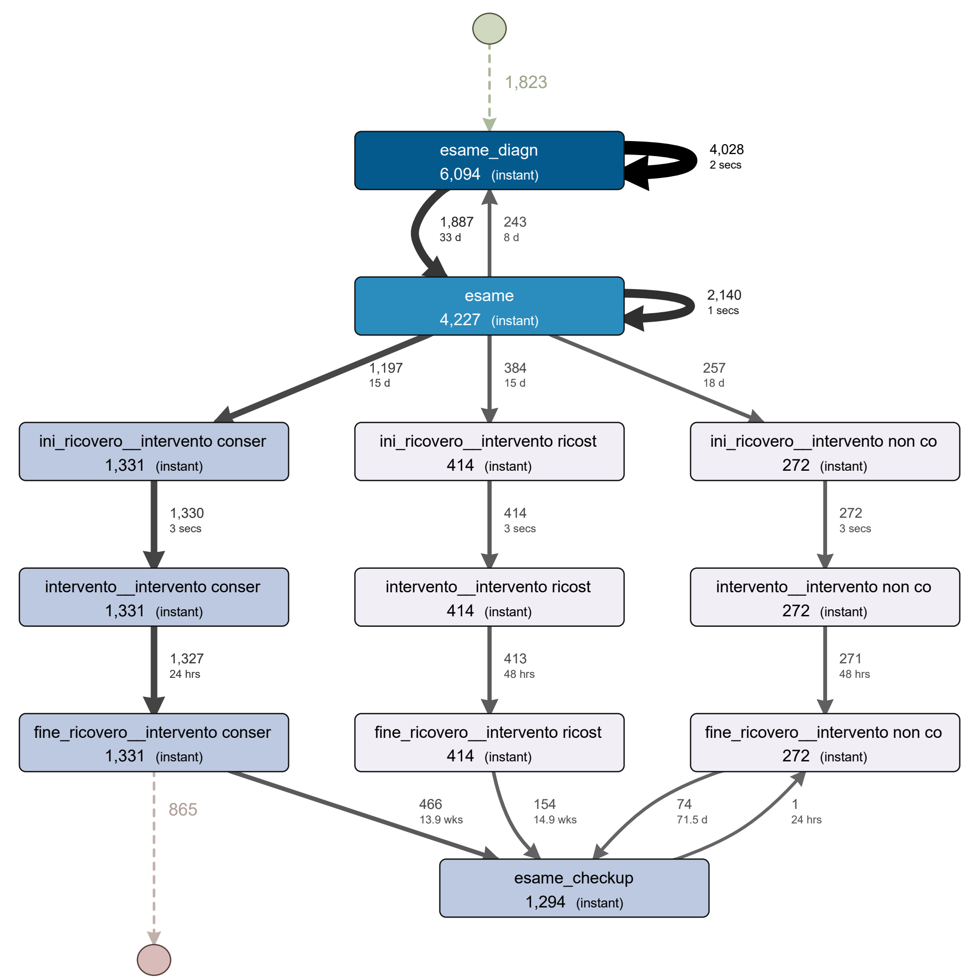
Enhancing quality and decision-making for care pathways: An application of process mining in cancer care

Francesca Ferré, Chiara Seghieri, Sima Sarv Ahrabi, Andrea Burattin, Andrea Vandin

Published: January 7, 2026 https://doi.org/10.1371/journal.pone.0339788

PLOS One





We will see a simple example (purchasing process)

However, PM has been applied to several domains, e.g., the **security** one

White-box validation of quantitative dynamic product lines by statistical model checking and process mining

Roberto Casaluce[a], Andrea Burattin[b], Francesca Chiaromonte[a,c], Alberto Lluch Lafuente[b], Andrea Vandin[a,b,*]

[a]Institute of Economics and EMbeDS, Sant'Anna School of Advanced Studies, Pisa, Italy.
[b]DTU Technical University of Denmark, Lyngby, Denmark.
[c]Dept. of Statistics and Huck Institutes of the Life Sciences, Penn State University, USA

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Journal of Systems and Software

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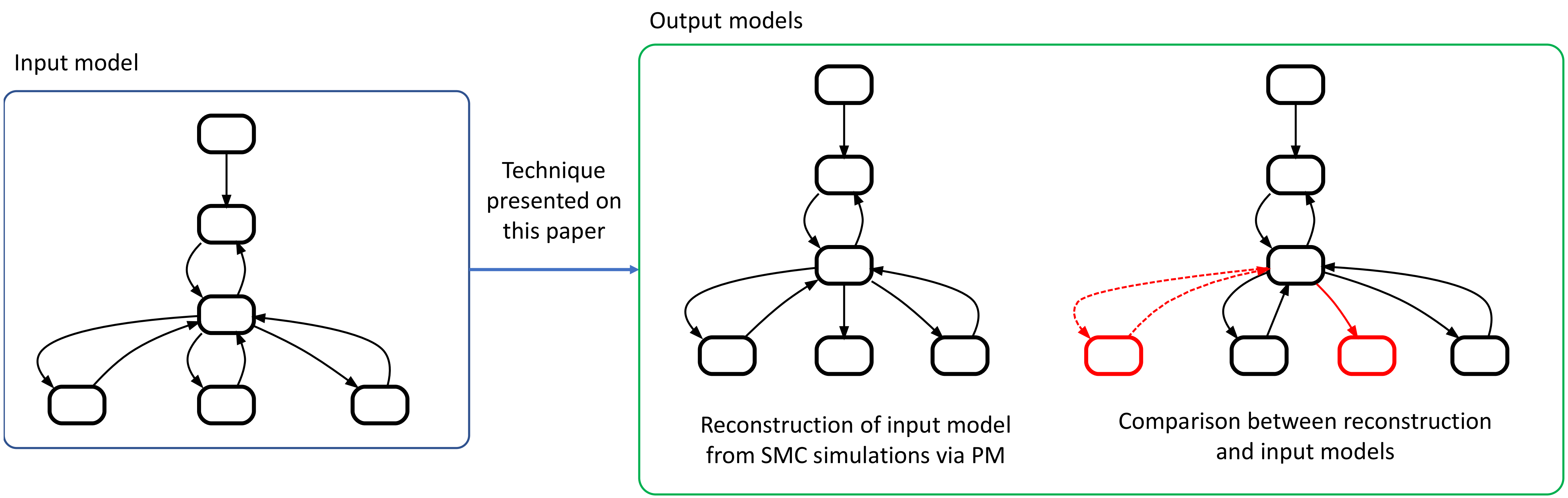
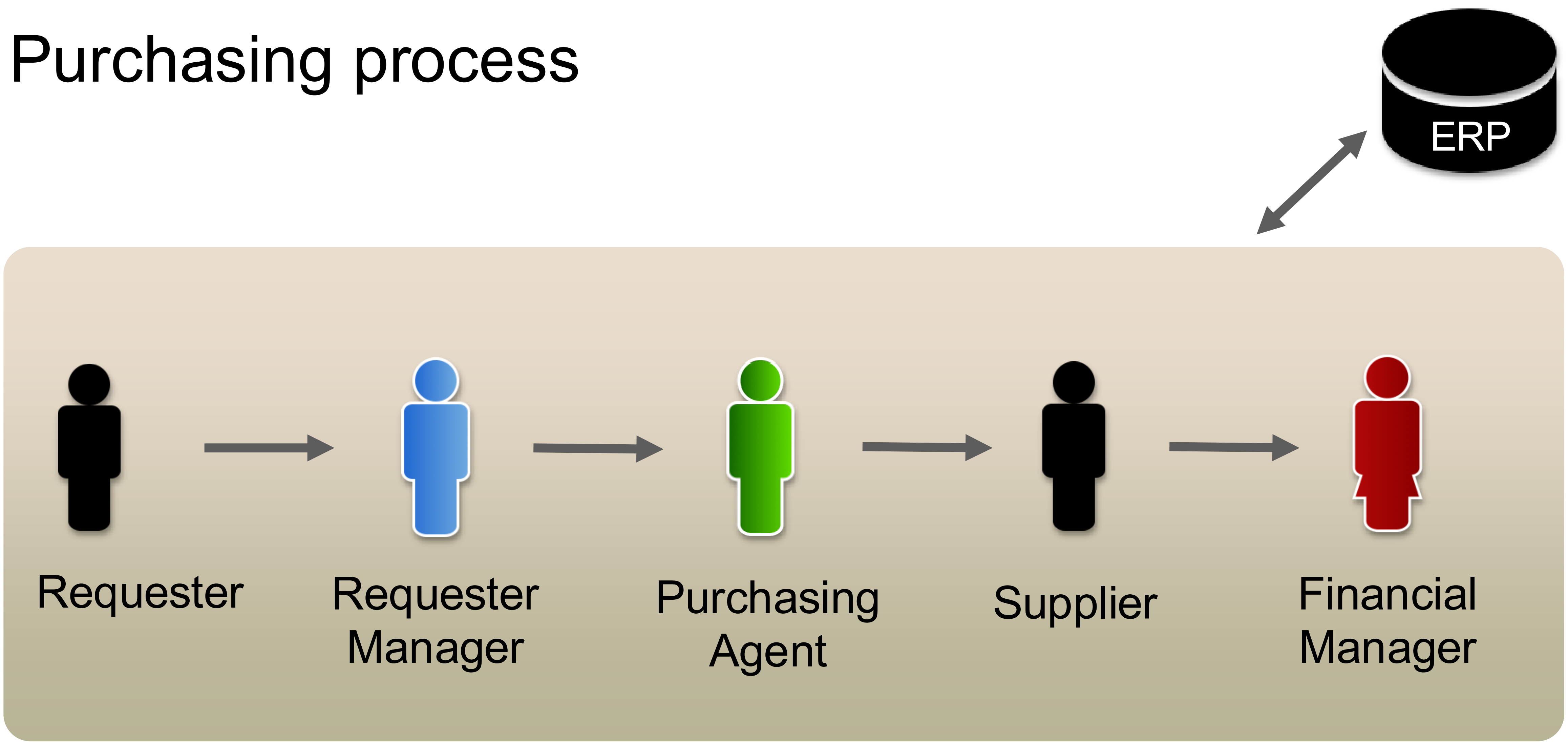


Outline

0. What is Process mining?
1. **Example Scenario**
2. Roadmap
3. Hands-on Session
4. Take-away Points

Example Scenario

Purchasing process



Problems

1. Inefficient operations
2. Need to demonstrate compliance
3. Complaints about process duration

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Analysis Goals

1. Understand the process in detail
2. Check whether there are deviations from the payment guidelines
3. Control performance targets
   - Complete process in at most 21 days

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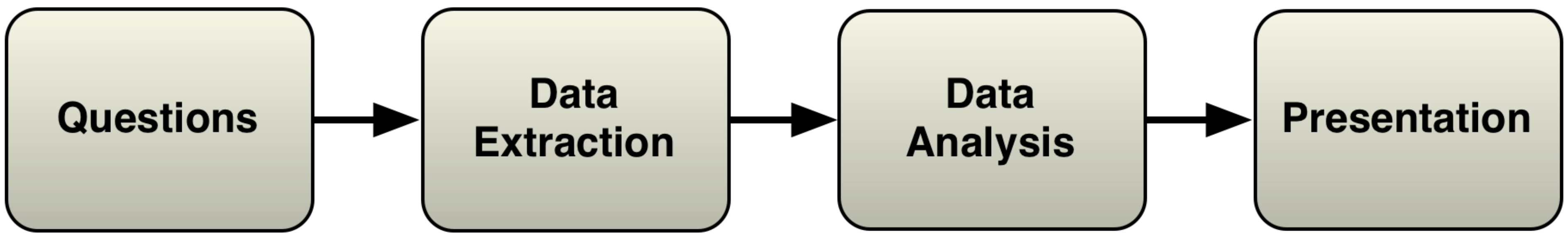
Roadmap



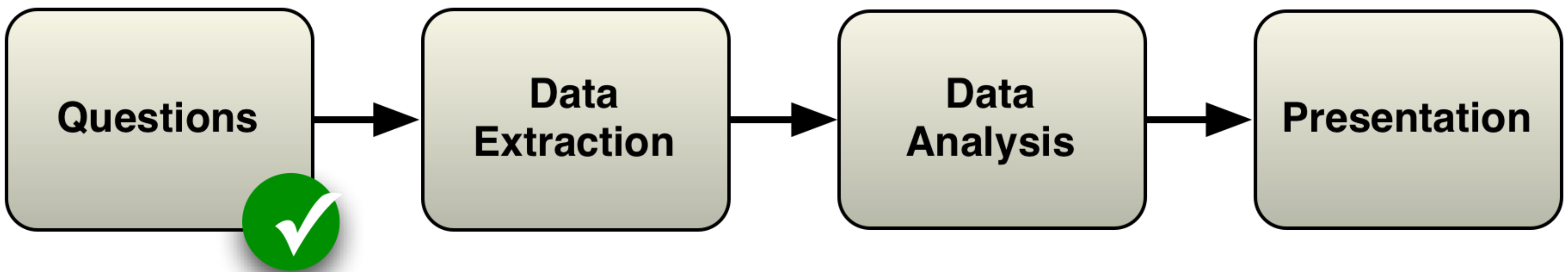
- Determine questions
- Process scope
- Which IT systems

- Via DB administrator
- CSV file or database extract

- Extract 'As-is' process
- Answer questions

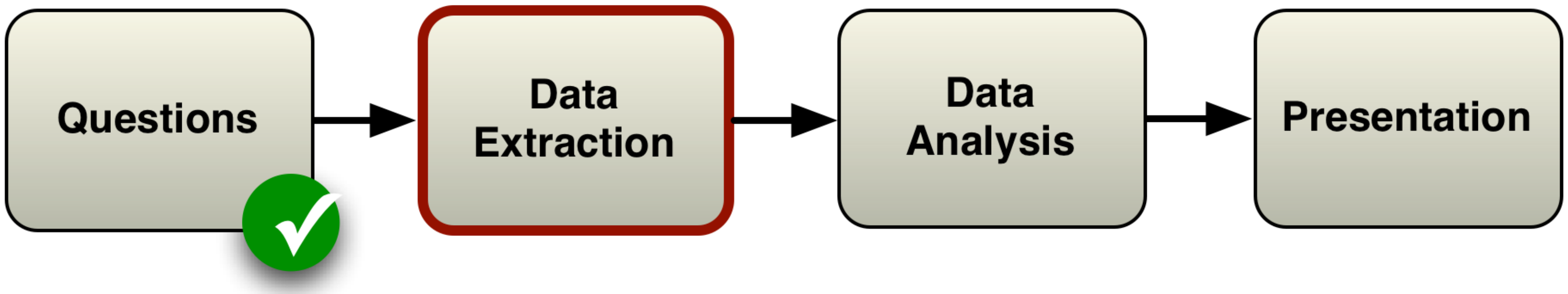
- Present results (e.g., report, presentation, workshop etc.)

Roadmap



1. How does the process actually look like?
2. Are there deviations from the prescribed process?
3. Do we meet the performance targets?

Roadmap



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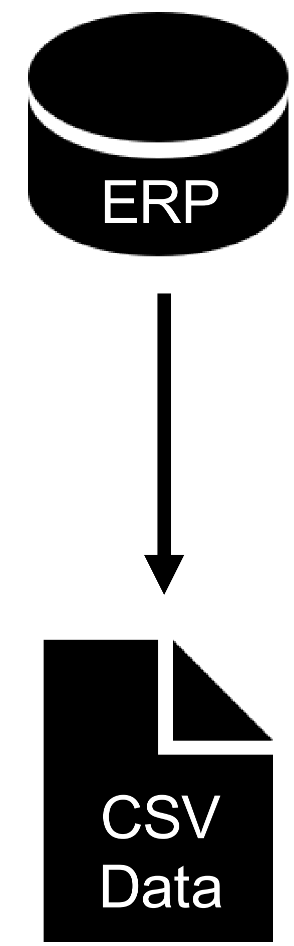
Data Extraction

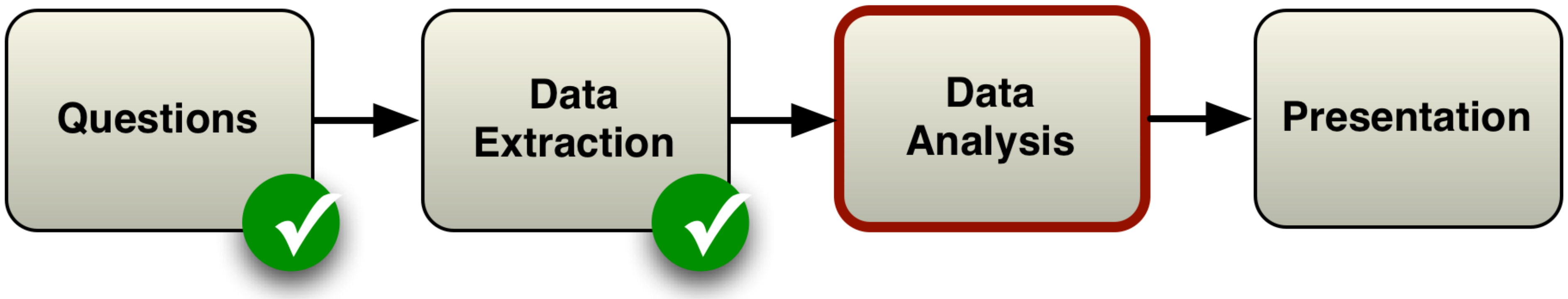
IT staff extracts history logs (from the Enterprise Resource Planning system, ERP) in a CSV file



Such CSV file is the starting point for our tutorial

- Please download it from our wiki (Slides page)
- Or here: PurchaseExample.csv

Roadmap



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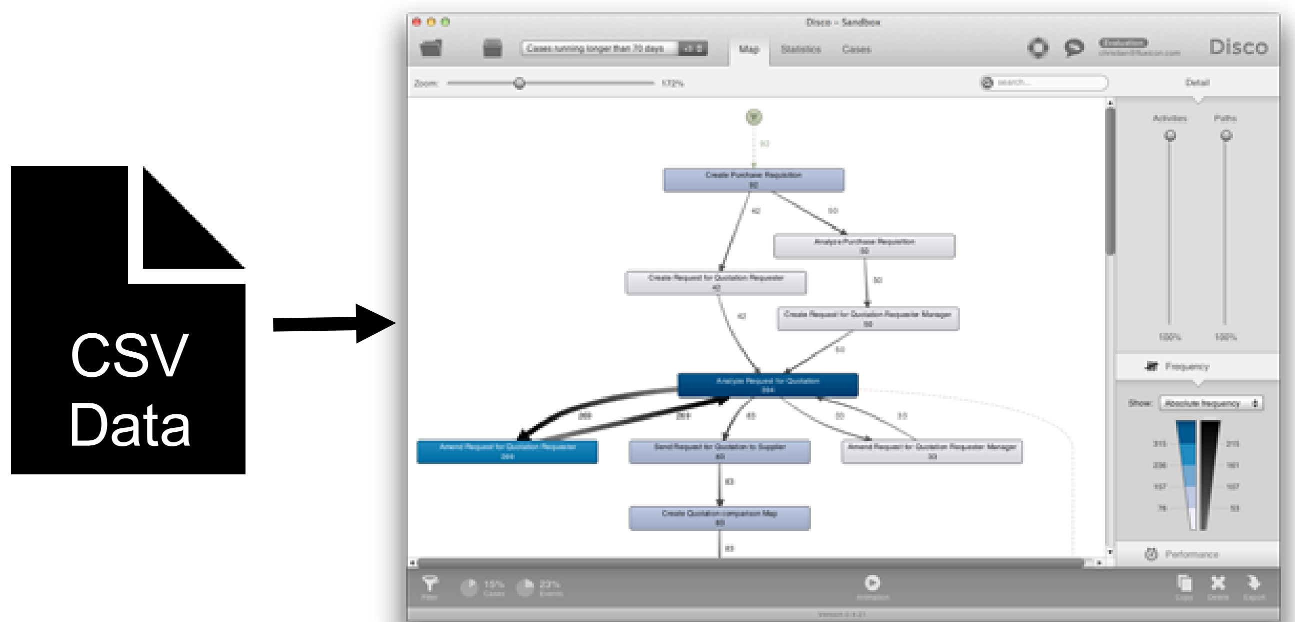
Data Analysis

We use the process mining tool Disco to perform the data analysis



Download from fluxicon.com/disco

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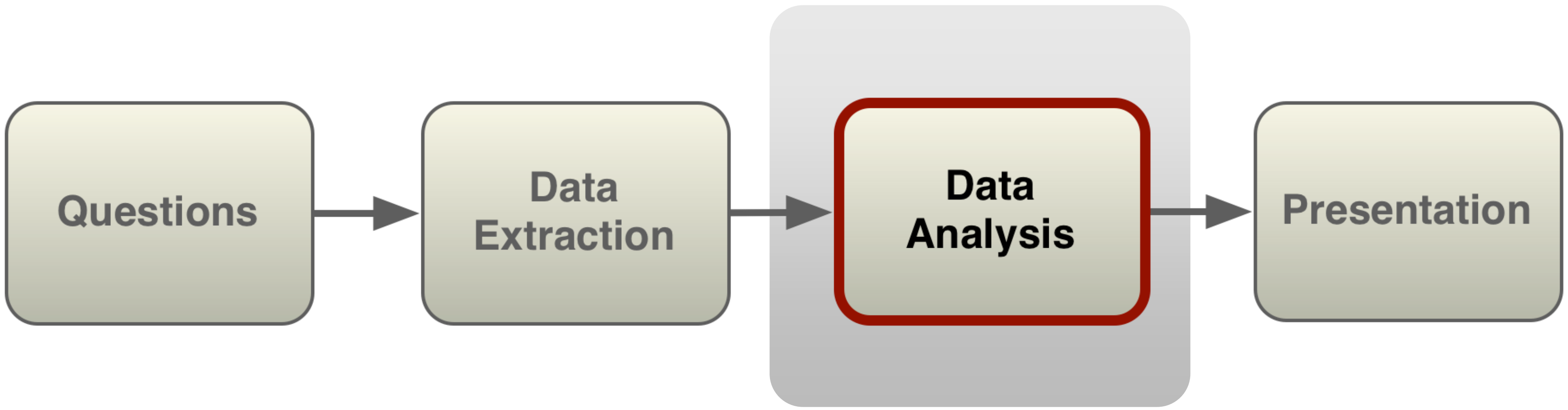
Roadmap



Focus of this session

Outline

0. What is Process mining?
1. Example Scenario
2. Roadmap
3. **Hands-on Session**
4. Take-away Points

Hands-on Session

Let's get started!

Step 1 - Inspect Data

Open `PurchasingExample.csv` file in python/pandas (or Excel) and inspect its contents

- Every row corresponds to one event
- You find information on Case IDs, Activities, Start and end times, Resources, Roles

If using excel…

PurchasingExample.csv

	A	B	C	D	E	F
1	Case ID	Start Timestamp	Complete Timestamp	Activity	Resource	Role
2	1	2011/01/01 00:00:00.000	2011/01/01 00:37:00.000	Create Purchase Requisition	Kim Passa	Requester
3	2	2011/01/01 00:16:00.000	2011/01/01 00:29:00.000	Create Purchase Requisition	Immanuel Karagianni	Requester
4	3	2011/01/01 02:23:00.000	2011/01/01 03:03:00.000	Create Purchase Requisition	Kim Passa	Requester
5	1	2011/01/01 05:37:00.000	2011/01/01 05:45:00.000	Create Request for Quotation	Kim Passa	Requester
6	1	2011/01/01 06:41:00.000	2011/01/01 06:55:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
7	2	2011/01/01 08:16:00.000	2011/01/01 08:26:00.000	Create Request for Quotation	Alberto Duport	Requester
8	4	2011/01/01 08:39:00.000	2011/01/01 09:00:00.000	Create Purchase Requisition	Fjodor Kowalski	Requester
9	2	2011/01/01 09:34:00.000	2011/01/01 09:38:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
10	5	2011/01/01 09:49:00.000	2011/01/01 10:35:00.000	Create Purchase Requisition	Esmana Liubiata	Requester
11	2	2011/01/01 10:16:00.000	2011/01/01 10:21:00.000	Amend Request for Quotation	Christian Francois	Requester
12	2	2011/01/01 11:15:00.000	2011/01/01 11:48:00.000	Analyze Request for Quotation	Magdalena Predutta	Purchasing Agent
13	6	2011/01/01 11:20:00.000	2011/01/01 11:37:00.000	Create Purchase Requisition	Christian Francois	Requester
14	1	2011/01/01 11:43:00.000	2011/01/01 12:09:00.000	Send Request for Quotation to Supplier	Karel de Groot	Purchasing Agent
15	1	2011/01/01 12:32:00.000	2011/01/01 16:03:00.000	Create Quotation comparison Map	Magdalena Predutta	Purchasing Agent
16	2	2011/01/01 12:33:00.000	2011/01/01 12:39:00.000	Amend Request for Quotation	Esmana Liubiata	Requester
17	2	2011/01/01 13:28:00.000	2011/01/01 13:38:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
18	7	2011/01/01 14:05:00.000	2011/01/01 15:00:00.000	Create Purchase Requisition	Esmana Liubiata	Requester
19	8	2011/01/01 14:27:00.000	2011/01/01 15:17:00.000	Create Purchase Requisition	Fjodor Kowalski	Requester
20	2	2011/01/01 15:18:00.000	2011/01/01 15:40:00.000	Send Request for Quotation to Supplier	Francois de Perrier	Purchasing Agent
21	2	2011/01/01 15:55:00.000	2011/01/01 16:43:00.000	Create Quotation comparison Map	Karel de Groot	Purchasing Agent
22	9	2011/01/01 16:17:00.000	2011/01/01 16:34:00.000	Create Purchase Requisition	Tesca Lobes	Requester
23	6	2011/01/01 17:32:00.000	2011/01/01 17:45:00.000	Create Request for Quotation	Alberto Duport	Requester
24	8	2011/01/01 18:00:00.000	2011/01/01 18:07:00.000	Create Request for Quotation	Tesca Lobes	Requester
25	6	2011/01/01 18:39:00.000	2011/01/01 18:55:00.000	Analyze Request for Quotation	Magdalena Predutta	Purchasing Agent
26	4	2011/01/01 18:45:00.000	2011/01/01 18:51:00.000	Analyze Purchase Requisition	Maris Freeman	Requester Manager
27	4	2011/01/01 18:56:00.000	2011/01/01 18:58:00.000	Create Request for Quotation	Heinz Gutschmidt	Requester Manager
28	8	2011/01/01 19:04:00.000	2011/01/01 19:27:00.000	Analyze Request for Quotation	Francois de Perrier	Purchasing Agent
29	6	2011/01/01 19:47:00.000	2011/01/01 19:55:00.000	Amend Request for Quotation	Penn Osterwalder	Requester
30	4	2011/01/01 19:58:00.000	2011/01/01 20:19:00.000	Analyze Request for Quotation	Francois de Perrier	Purchasing Agent
31	8	2011/01/01 20:21:00.000	2011/01/01 20:34:00.000	Amend Request for Quotation	Tesca Lobes	Requester
32	6	2011/01/01 20:55:00.000	2011/01/01 21:28:00.000	Analyze Request for Quotation	Francois de Perrier	Purchasing Agent
33	4	2011/01/01 21:06:00.000	2011/01/01 21:14:00.000	Amend Request for Quotation	Nico Ojenbeer	Requester
34	8	2011/01/01 21:35:00.000	2011/01/01 22:01:00.000	Analyze Request for Quotation	Magdalena Predutta	Purchasing Agent
35	4	2011/01/01 22:01:00.000	2011/01/01 22:24:00.000	Analyze Request for Quotation	Francois de Perrier	Purchasing Agent
36	6	2011/01/01 22:13:00.000	2011/01/01 22:28:00.000	Amend Request for Quotation	Anne Olwada	Requester

If using pandas (python)…

```
logs.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 9119 entries, 0 to 9118
Data columns (total 6 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Case ID             9119 non-null   int64
 1   Start Timestamp     9119 non-null   datetime64[ns]
 2   Complete Timestamp  9119 non-null   datetime64[ns]
 3   Activity            9119 non-null   object
 4   Resource            9119 non-null   object
 5   Role                9119 non-null   object
dtypes: datetime64[ns](2), int64(1), object(3)
memory usage: 498.7+ KB
```

```
logs
```

	Case ID	Start Timestamp	Complete Timestamp	Activity	Resource	Role
0	1	2011-01-01 00:00:00	2011-01-01 00:37:00	Create Purchase Requisition	Kim Passa	Requester
1	2	2011-01-01 00:16:00	2011-01-01 00:29:00	Create Purchase Requisition	Immanuel Karagianni	Requester
2	3	2011-01-01 02:23:00	2011-01-01 03:03:00	Create Purchase Requisition	Kim Passa	Requester
3	1	2011-01-01 05:37:00	2011-01-01 05:45:00	Create Request for Quotation	Kim Passa	Requester
4	1	2011-01-01 06:41:00	2011-01-01 06:55:00	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
...	...	...	...	...	...	...
9114	1284	2011-10-14 13:53:00	2011-10-14 14:07:00	Pay Invoice	Pedro Alvares	Financial Manager
9115	1448	2011-10-14 13:56:00	2011-10-14 14:24:00	Settle Dispute With Supplier	Karalda Nimwada	Financial Manager
9116	1941	2011-10-14 14:05:00	2011-10-14 14:18:00	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
9117	1448	2011-10-14 14:24:00	2011-10-14 14:24:00	Authorize Supplier's Invoice payment	Karalda Nimwada	Financial Manager
9118	1448	2011-10-14 15:16:00	2011-10-14 15:31:00	Pay Invoice	Karalda Nimwada	Financial Manager

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 5   Role                9119 non-null   object
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memory usage: 498.7+ KB
```

```
logs[logs["Case ID"]==1]
```

	Case ID	Start Timestamp	Complete Timestamp	Activity	Resource	Role
0	1	2011-01-01 00:00:00	2011-01-01 00:37:00	Create Purchase Requisition	Kim Passa	Requester
3	1	2011-01-01 05:37:00	2011-01-01 05:45:00	Create Request for Quotation	Kim Passa	Requester
4	1	2011-01-01 06:41:00	2011-01-01 06:55:00	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
12	1	2011-01-01 11:43:00	2011-01-01 12:09:00	Send Request for Quotation to Supplier	Karel de Groot	Purchasing Agent
13	1	2011-01-01 12:32:00	2011-01-01 16:03:00	Create Quotation comparison Map	Magdalena Predutta	Purchasing Agent
36	1	2011-01-01 22:44:00	2011-01-01 23:13:00	Analyze Quotation Comparison Map	Immanuel Karagianni	Requester
37	1	2011-01-01 23:13:00	2011-01-01 23:13:00	Choose best option	Tesca Lobes	Requester
42	1	2011-01-02 01:22:00	2011-01-02 09:20:00	Settle Conditions With Supplier	Francois de Perrier	Purchasing Agent
60	1	2011-01-02 09:58:00	2011-01-02 10:10:00	Create Purchase Order	Karel de Groot	Purchasing Agent
68	1	2011-01-02 14:09:00	2011-01-02 14:43:00	Confirm Purchase Order	Sean Manney	Supplier
97	1	2011-01-02 20:49:00	2011-01-03 03:37:00	Deliver Goods Services	Sean Manney	Supplier
137	1	2011-01-03 11:20:00	2011-01-03 11:21:00	Release Purchase Order	Elvira Lores	Requester
176	1	2011-01-03 19:09:00	2011-01-03 19:10:00	Approve Purchase Order for payment	Karel de Groot	Purchasing Agent
221	1	2011-01-04 00:54:00	2011-01-04 00:54:00	Send Invoice	Kiu Kan	Supplier
276	1	2011-01-04 15:08:00	2011-01-04 15:13:00	Release Supplier's Invoice	Karalda Nimwada	Financial Manager
277	1	2011-01-04 15:13:00	2011-01-04 15:13:00	Authorize Supplier's Invoice payment	Karalda Nimwada	Financial Manager
278	1	2011-01-04 15:22:00	2011-01-04 15:31:00	Pay Invoice	Pedro Alvares	Financial Manager

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 1   Start Timestamp     9119 non-null   datetime64[ns]
 2   Complete Timestamp  9119 non-null   datetime64[ns]
```

```
logs[logs["Case ID"]==2]
```

	Case ID	Start Timestamp	Complete Timestamp	Activity	Resource	Role
1	2	2011-01-01 00:16:00	2011-01-01 00:29:00	Create Purchase Requisition	Immanuel Karagianni	Requester
5	2	2011-01-01 08:16:00	2011-01-01 08:26:00	Create Request for Quotation	Alberto Duport	Requester
7	2	2011-01-01 09:34:00	2011-01-01 09:38:00	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
9	2	2011-01-01 10:16:00	2011-01-01 10:21:00	Amend Request for Quotation	Christian Francois	Requester
10	2	2011-01-01 11:15:00	2011-01-01 11:48:00	Analyze Request for Quotation	Magdalena Predutta	Purchasing Agent
14	2	2011-01-01 12:33:00	2011-01-01 12:39:00	Amend Request for Quotation	Esmana Liubiata	Requester
15	2	2011-01-01 13:28:00	2011-01-01 13:38:00	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
18	2	2011-01-01 15:18:00	2011-01-01 15:40:00	Send Request for Quotation to Supplier	Francois de Perrier	Purchasing Agent
19	2	2011-01-01 15:55:00	2011-01-01 16:43:00	Create Quotation comparison Map	Karel de Groot	Purchasing Agent
39	2	2011-01-01 23:33:00	2011-01-01 23:44:00	Analyze Quotation Comparison Map	Esmana Liubiata	Requester
40	2	2011-01-01 23:44:00	2011-01-01 23:44:00	Choose best option	Fjodor Kowalski	Requester
47	2	2011-01-02 02:15:00	2011-01-02 15:07:00	Settle Conditions With Supplier	Magdalena Predutta	Purchasing Agent
77	2	2011-01-02 16:21:00	2011-01-02 16:31:00	Create Purchase Order	Magdalena Predutta	Purchasing Agent
94	2	2011-01-02 20:23:00	2011-01-02 20:29:00	Confirm Purchase Order	Esmeralda Clay	Supplier
123	2	2011-01-03 03:15:00	2011-01-04 14:53:00	Deliver Goods Services	Esmeralda Clay	Supplier
312	2	2011-01-04 23:01:00	2011-01-04 23:02:00	Release Purchase Order	Elvira Lores	Requester
342	2	2011-01-05 09:03:00	2011-01-05 09:04:00	Approve Purchase Order for payment	Karel de Groot	Purchasing Agent
367	2	2011-01-05 14:50:00	2011-01-05 14:50:00	Send Invoice	Esmeralda Clay	Supplier
409	2	2011-01-06 05:04:00	2011-01-06 05:07:00	Release Supplier's Invoice	Pedro Alvares	Financial Manager
411	2	2011-01-06 05:38:00	2011-01-06 05:58:00	Settle Dispute With Supplier	Karalda Nimwada	Financial Manager
413	2	2011-01-06 05:58:00	2011-01-06 05:58:00	Authorize Supplier's Invoice payment	Pedro Alvares	Financial Manager
415	2	2011-01-06 06:21:00	2011-01-06 06:32:00	Pay Invoice	Karalda Nimwada	Financial Manager

Step 2 - Import Data

Load **`PurchasingExample.csv`** in Disco

Assign columns as follows:

- Case ID → Case ID
- Start and Complete Timestamp → Timestamp
- Activity → Activity
- Resource → Resource
- Role → Other

Click ‘Start import’

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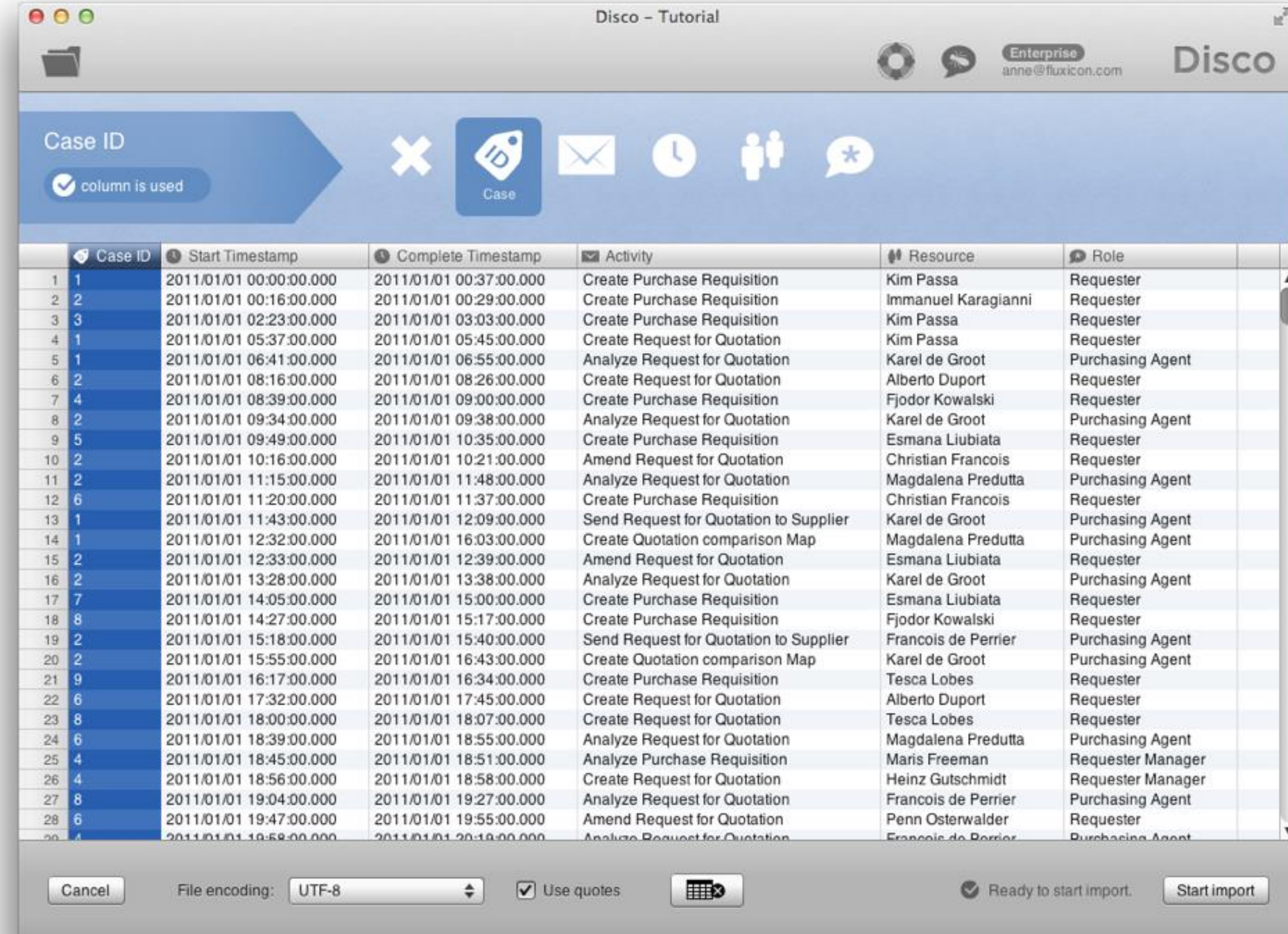


	Case ID	Start Timestamp	Complete Timestamp	Activity	Resource	Role
1	1	2011/01/01 00:00:00.000	2011/01/01 00:37:00.000	Create Purchase Requisition	Kim Passa	Requester
2	2	2011/01/01 00:16:00.000	2011/01/01 00:29:00.000	Create Purchase Requisition	Immanuel Karagianni	Requester
3	3	2011/01/01 02:23:00.000	2011/01/01 03:03:00.000	Create Purchase Requisition	Kim Passa	Requester
4	1	2011/01/01 05:37:00.000	2011/01/01 05:45:00.000	Create Request for Quotation	Kim Passa	Requester
5	1	2011/01/01 06:41:00.000	2011/01/01 06:55:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
6	2	2011/01/01 08:16:00.000	2011/01/01 08:26:00.000	Create Request for Quotation	Alberto Duport	Requester
7	4	2011/01/01 08:39:00.000	2011/01/01 09:00:00.000	Create Purchase Requisition	Fjodor Kowalski	Requester
8	2	2011/01/01 09:34:00.000	2011/01/01 09:38:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
9	5	2011/01/01 09:49:00.000	2011/01/01 10:35:00.000	Create Purchase Requisition	Esmana Liubiata	Requester
10	2	2011/01/01 10:16:00.000	2011/01/01 10:21:00.000	Amend Request for Quotation	Christian Francois	Requester
11	2	2011/01/01 11:15:00.000	2011/01/01 11:48:00.000	Analyze Request for Quotation	Magdalena Predutta	Purchasing Agent
12	6	2011/01/01 11:20:00.000	2011/01/01 11:37:00.000	Create Purchase Requisition	Christian Francois	Requester
13	1	2011/01/01 11:43:00.000	2011/01/01 12:09:00.000	Send Request for Quotation to Supplier	Karel de Groot	Purchasing Agent
14	1	2011/01/01 12:32:00.000	2011/01/01 16:03:00.000	Create Quotation comparison Map	Magdalena Predutta	Purchasing Agent
15	2	2011/01/01 12:33:00.000	2011/01/01 12:39:00.000	Amend Request for Quotation	Esmana Liubiata	Requester
16	2	2011/01/01 13:28:00.000	2011/01/01 13:38:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
17	7	2011/01/01 14:05:00.000	2011/01/01 15:00:00.000	Create Purchase Requisition	Esmana Liubiata	Requester
18	8	2011/01/01 14:27:00.000	2011/01/01 15:17:00.000	Create Purchase Requisition	Fjodor Kowalski	Requester
19	2	2011/01/01 15:18:00.000	2011/01/01 15:40:00.000	Send Request for Quotation to Supplier	Francois de Perrier	Purchasing Agent
20	2	2011/01/01 15:55:00.000	2011/01/01 16:43:00.000	Create Quotation comparison Map	Karel de Groot	Purchasing Agent
21	9	2011/01/01 16:17:00.000	2011/01/01 16:34:00.000	Create Purchase Requisition	Tesca Lobes	Requester
22	6	2011/01/01 17:32:00.000	2011/01/01 17:45:00.000	Create Request for Quotation	Alberto Duport	Requester
23	8	2011/01/01 18:00:00.000	2011/01/01 18:07:00.000	Create Request for Quotation	Tesca Lobes	Requester
24	6	2011/01/01 18:39:00.000	2011/01/01 18:55:00.000	Analyze Request for Quotation	Magdalena Predutta	Purchasing Agent
25	4	2011/01/01 18:45:00.000	2011/01/01 18:51:00.000	Analyze Purchase Requisition	Maris Freeman	Requester Manager
26	4	2011/01/01 18:56:00.000	2011/01/01 18:58:00.000	Create Request for Quotation	Heinz Gutschmidt	Requester Manager
27	8	2011/01/01 19:04:00.000	2011/01/01 19:27:00.000	Analyze Request for Quotation	Francois de Perrier	Purchasing Agent
28	6	2011/01/01 19:47:00.000	2011/01/01 19:55:00.000	Amend Request for Quotation	Penn Osterwalder	Requester

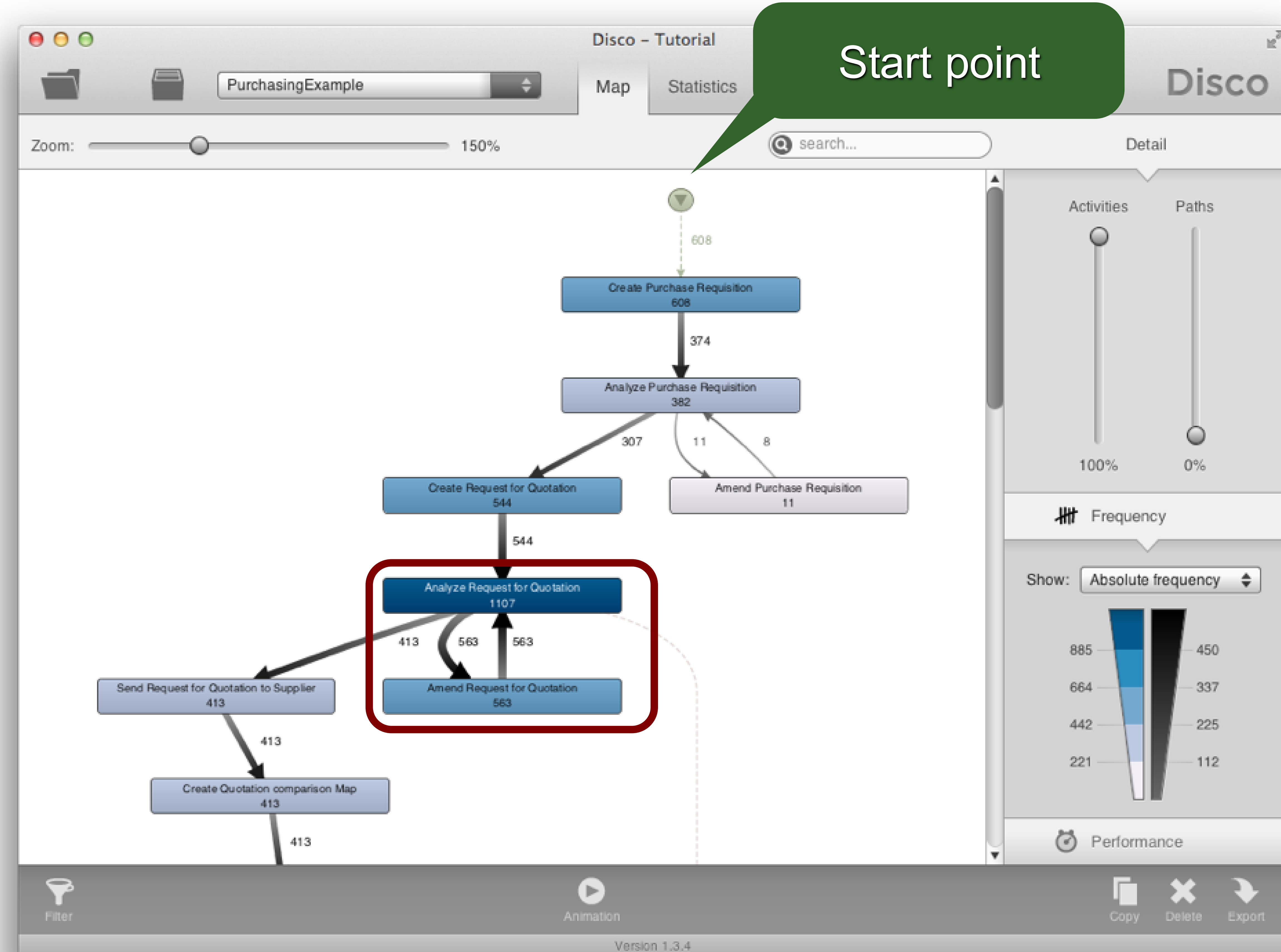
Step 3 - Inspect Process

Look at the resulting process model

- Numbers in rectangles are activity frequencies
- Number at arcs is frequency of connection

→ You see the main process flows

- All 608 cases start with activity ‘Create Purchase Requisition’
- Lots of changes were made (amendments)!



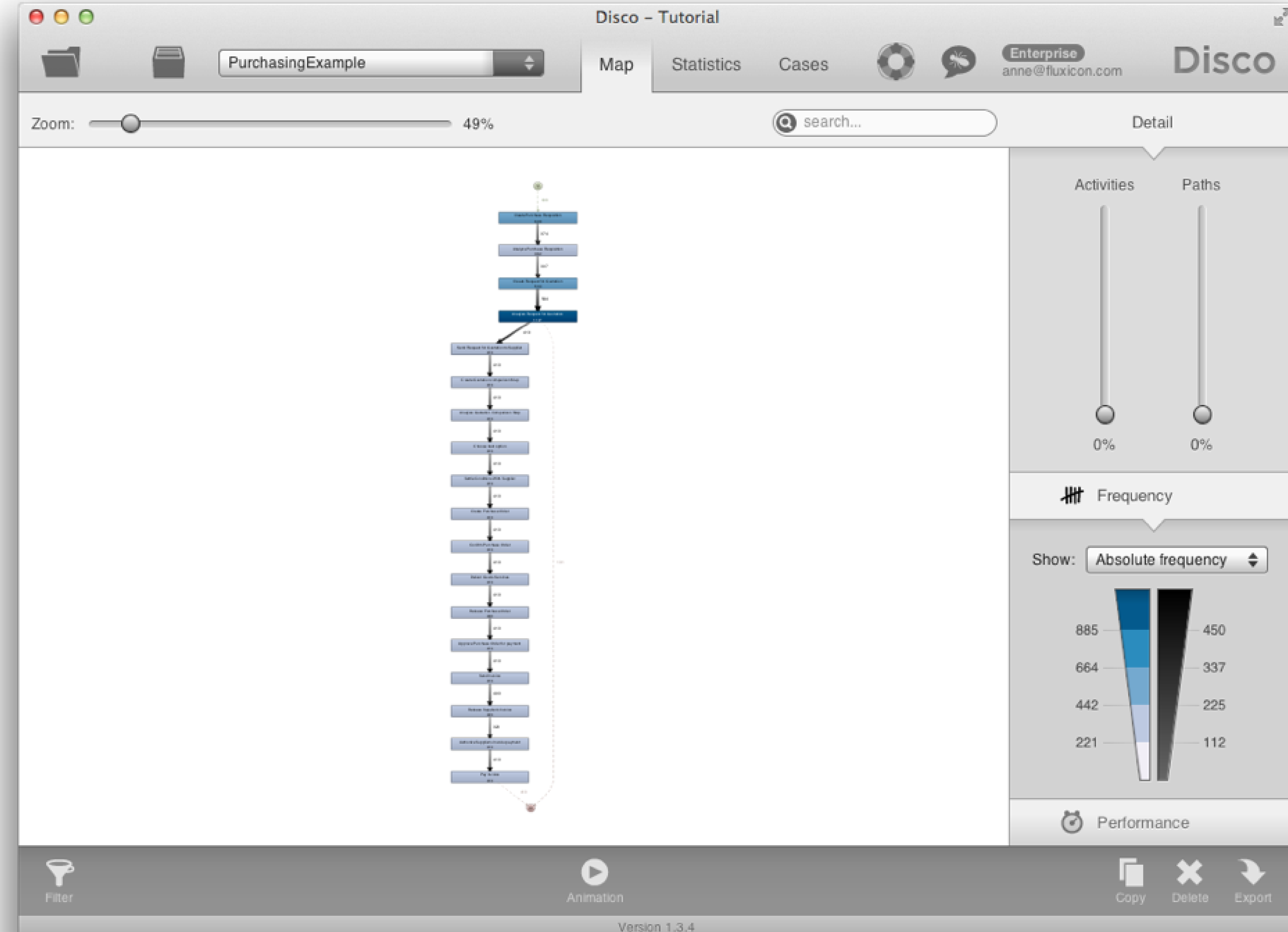
Start point
Disco - Tutorial
PurchasingExample
Map
Statistics
Disco
Zoom:
150%
search...
Detail
Activities
Paths
100%
0%
Frequency
Show:
Absolute frequency
885
664
442
221
450
337
225
112
Performance
608
Create Purchase Requisition
608
374
Analyze Purchase Requisition
382
307
11
8
Create Request for Quotation
544
Amend Purchase Requisition
11
544
Analyze Request for Quotation
1107
413
563
563
Send Request for Quotation to Supplier
413
Amend Request for Quotation
563
413
Create Quotation comparison Map
413
413
Filter
Animation
Copy
Delete
Export
Version 1.3.4

Step 3 - Inspect Process

It’s important to be able to adjust the level of detail for the process map

Move up the ‘Activities’ slider down to lowest position (0%)

- Only the activities from the most frequent process variant are shown


Disco – Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
49%
search...
Detail
Activities
Paths
0%
0%
Frequency
Show:
Absolute frequency
885
664
442
221
450
337
225
112
Performance
Filter
Animation
Copy
Delete
Export
Version 1.3.4

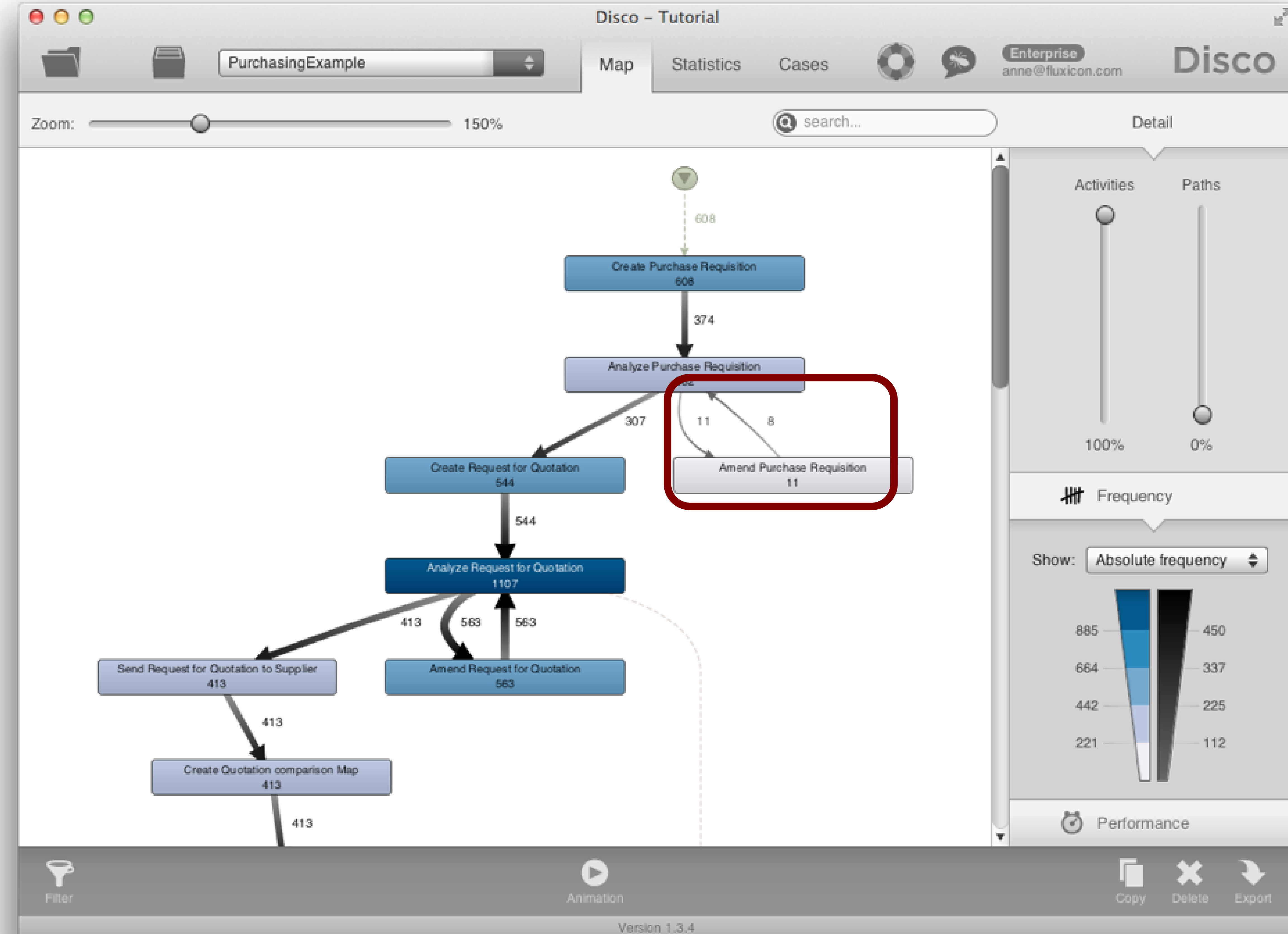
fluxicon

Step 3 - Inspect Process

Gradually move the ‘Activities’ slider up to 100% again until all activities are shown

- Even infrequent activities such as ‘Amend Purchase Requisition’ are shown

You’ll notice that 11 cases are flowing in to ‘Amend Purchase Requisition’ but only 8 are moving out - Where are the other 3?



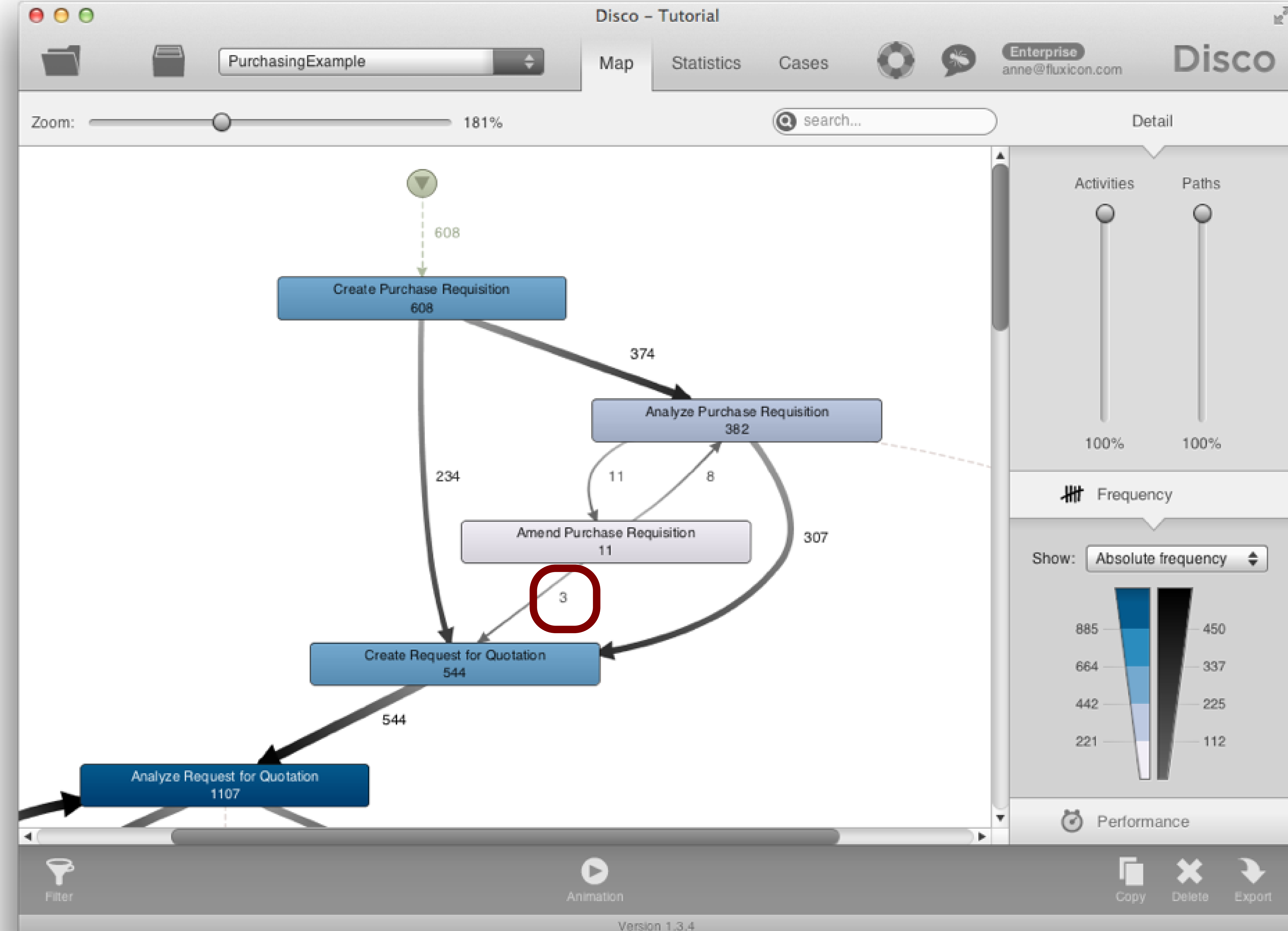
Disco – Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
150%
search...
Detail
Activities
Paths
100%
0%
608
Create Purchase Requisition
608
374
Analyze Purchase Requisition
307
11
8
Create Request for Quotation
544
Amend Purchase Requisition
11
544
Analyze Request for Quotation
1107
413
563
563
Send Request for Quotation to Supplier
413
Amend Request for Quotation
563
413
Create Quotation comparison Map
413
413
Frequency
Show:
Absolute frequency
885
664
442
221
450
337
225
112
Performance
Filter
Animation
Copy
Delete
Export
Version 1.3.4

Step 3 - Inspect Process

Move up the ‘Paths’ slider up to the top

You now see a 100% detailed picture of the executed process

- The 3 missing cases move from ‘Amend Purchase Requisition’ to ‘Create Request for Quotation’

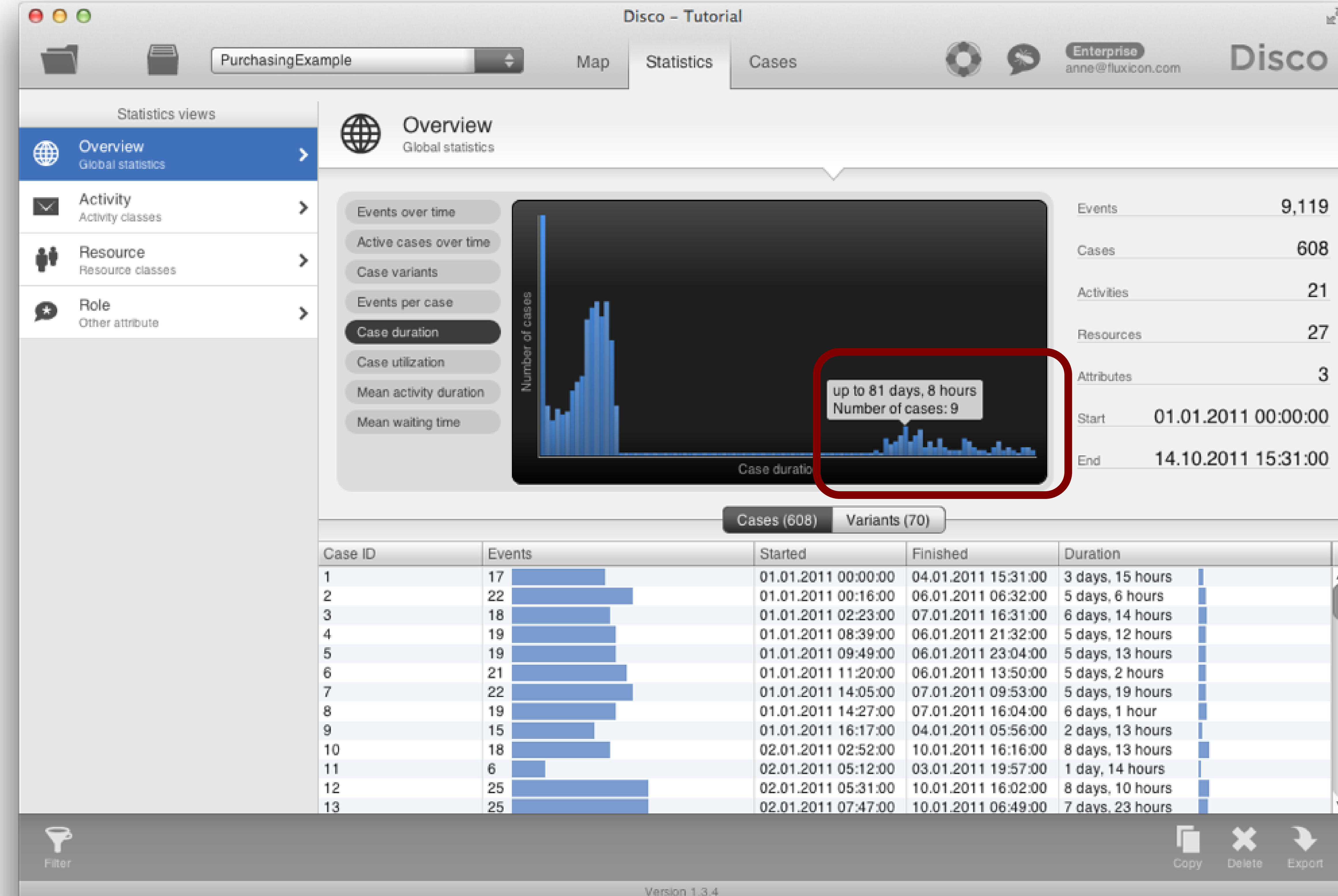

Disco – Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
181%
search...
Detail
Activities
Paths
100%
100%
Frequency
Show:
Absolute frequency
885
664
442
221
450
337
225
112
Performance
608
Create Purchase Requisition
608
374
Analyze Purchase Requisition
382
234
11
8
Amend Purchase Requisition
11
307
3
Create Request for Quotation
544
544
Analyze Request for Quotation
1107
Filter
Animation
Copy
Delete
Export
Version 1.3.4

Step 4 - Inspect Statistics

Look at ‘Statistics’ tab to see ‘Overview’ information about the event log

- 9,119 events were recorded for 608 cases
- Timeframe is January - October 2011

The ‘Case duration’ is typically up to 15 or 16 days, but some cases take very long (more than 70 or 80 days!)



Disco – Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Statistics views
Overview
Global statistics
Activity
Activity classes
Resource
Resource classes
Role
Other attribute
Overview
Global statistics
Events over time
Active cases over time
Case variants
Events per case
Case duration
Case utilization
Mean activity duration
Mean waiting time
Number of cases
up to 81 days, 8 hours
Number of cases: 9
Case duratio
Events 9,119
Cases 608
Activities 21
Resources 27
Attributes 3
Start 01.01.2011 00:00:00
End 14.10.2011 15:31:00
Cases (608)
Variants (70)
Case ID | Events | Started | Finished | Duration
1 | 17 | 01.01.2011 00:00:00 | 04.01.2011 15:31:00 | 3 days, 15 hours
2 | 22 | 01.01.2011 00:16:00 | 06.01.2011 06:32:00 | 5 days, 6 hours
3 | 18 | 01.01.2011 02:23:00 | 07.01.2011 16:31:00 | 6 days, 14 hours
4 | 19 | 01.01.2011 08:39:00 | 06.01.2011 21:32:00 | 5 days, 12 hours
5 | 19 | 01.01.2011 09:49:00 | 06.01.2011 23:04:00 | 5 days, 13 hours
6 | 21 | 01.01.2011 11:20:00 | 06.01.2011 13:50:00 | 5 days, 2 hours
7 | 22 | 01.01.2011 14:05:00 | 07.01.2011 09:53:00 | 5 days, 19 hours
8 | 19 | 01.01.2011 14:27:00 | 07.01.2011 16:04:00 | 6 days, 1 hour
9 | 15 | 01.01.2011 16:17:00 | 04.01.2011 05:56:00 | 2 days, 13 hours
10 | 18 | 02.01.2011 02:52:00 | 10.01.2011 16:16:00 | 8 days, 13 hours
11 | 6 | 02.01.2011 05:12:00 | 03.01.2011 19:57:00 | 1 day, 14 hours
12 | 25 | 02.01.2011 05:31:00 | 10.01.2011 16:02:00 | 8 days, 10 hours
13 | 25 | 02.01.2011 07:47:00 | 10.01.2011 06:49:00 | 7 days, 23 hours
Filter
Copy
Delete
Export
Version 1.3.4

fluxicon

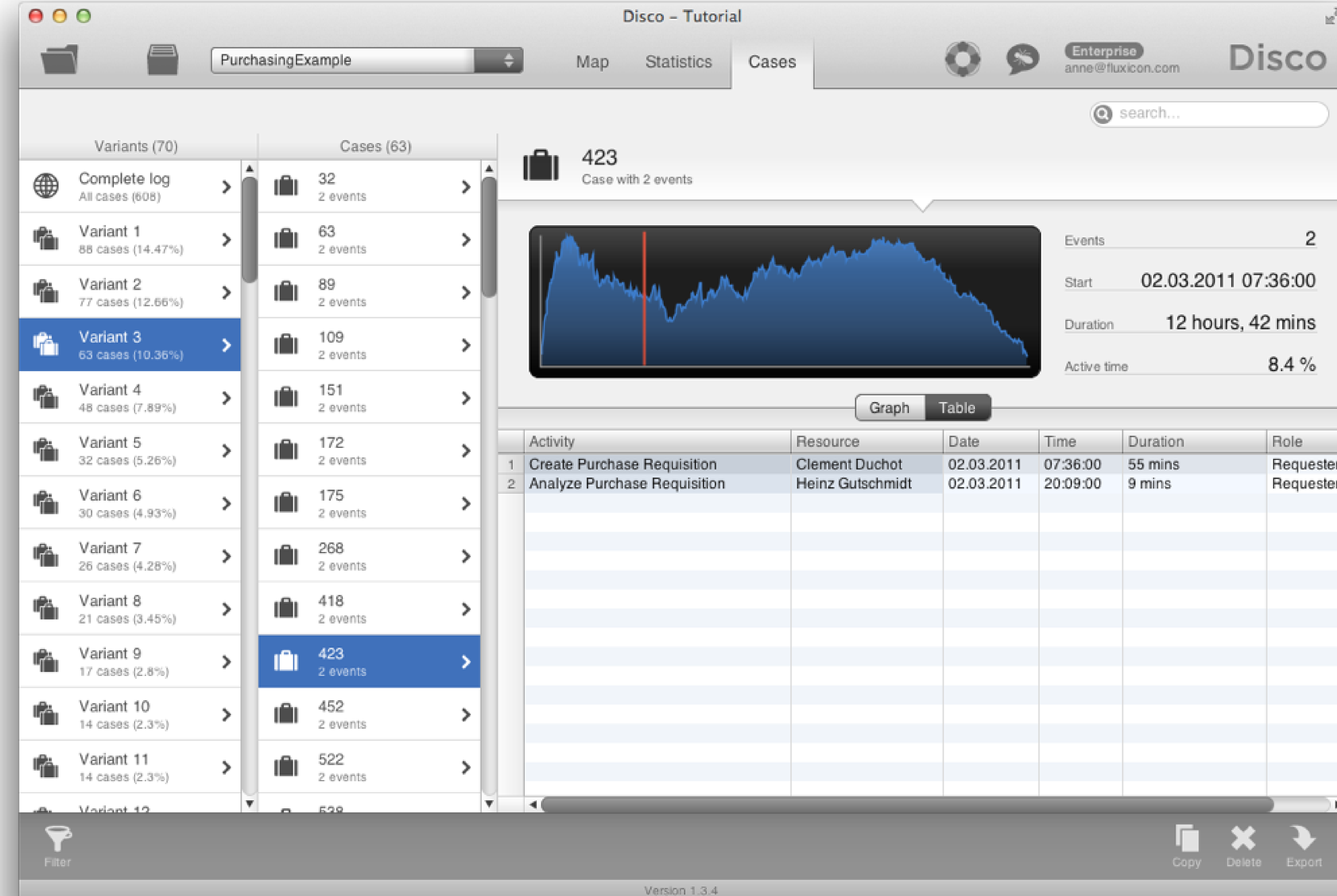
Step 5 - Inspect Cases

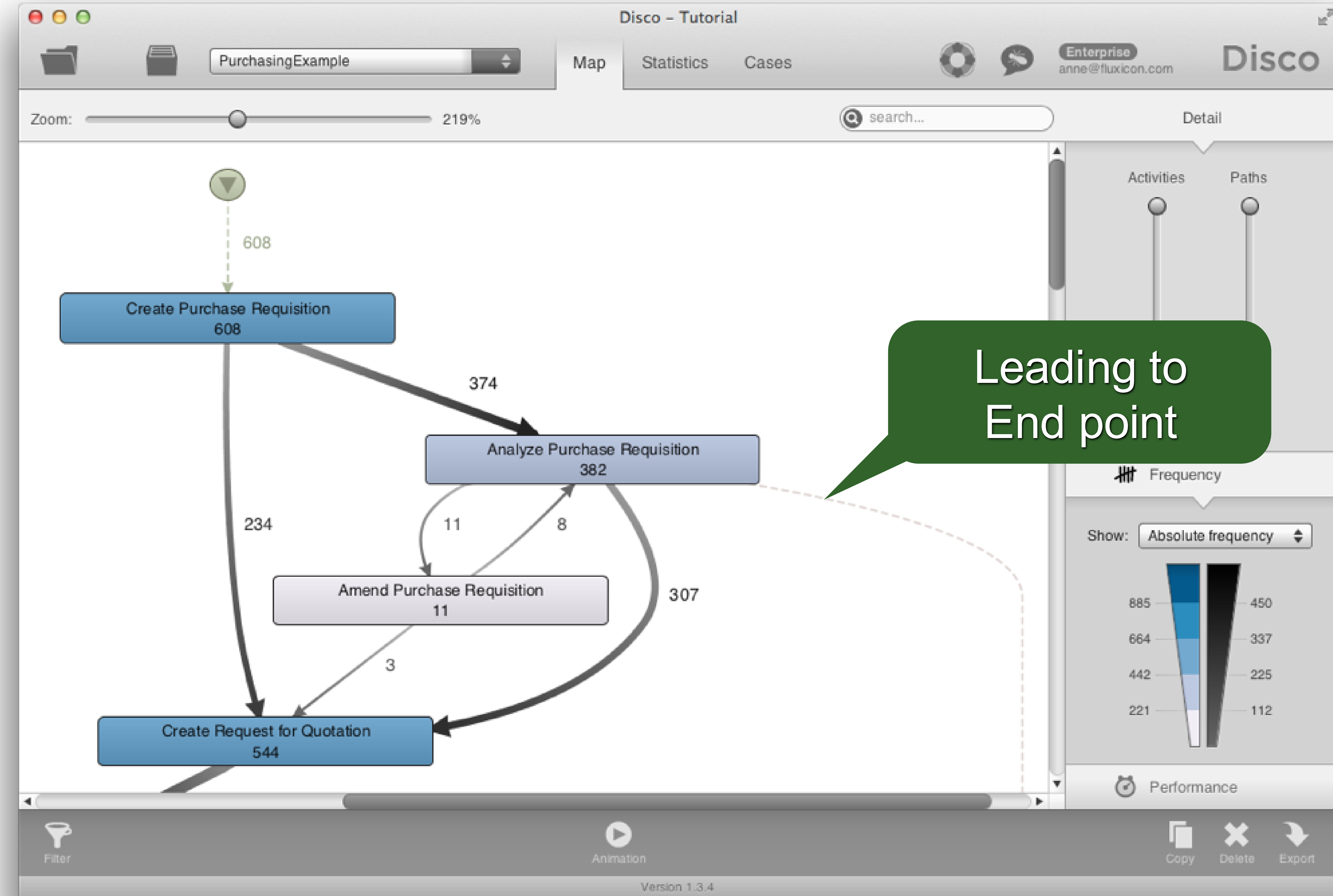
Select ‘Cases’ tab to inspect variants and individual service instances

- The third most frequent process variant ends after ‘Analyze Purchase Requisition’ (ca. 10.36% of all cases follow this pattern)

→ Why are so many requests abrupted?

- Do people not know what they can buy?
- We can find this back in the process map, too


Disco - Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
search...
Variants (70)
Cases (63)
Complete log
All cases (608)
Variant 1
88 cases (14.47%)
Variant 2
77 cases (12.66%)
Variant 3
63 cases (10.36%)
Variant 4
48 cases (7.89%)
Variant 5
32 cases (5.26%)
Variant 6
30 cases (4.93%)
Variant 7
26 cases (4.28%)
Variant 8
21 cases (3.45%)
Variant 9
17 cases (2.8%)
Variant 10
14 cases (2.3%)
Variant 11
14 cases (2.3%)
32 2 events
63 2 events
89 2 events
109 2 events
151 2 events
172 2 events
175 2 events
268 2 events
418 2 events
423 2 events
452 2 events
522 2 events
423
Case with 2 events
Events 2
Start 02.03.2011 07:36:00
Duration 12 hours, 42 mins
Active time 8.4 %
Graph
Table
Activity | Resource | Date | Time | Duration | Role
1 Create Purchase Requisition | Clement Duchot | 02.03.2011 | 07:36:00 | 55 mins | Requester
2 Analyze Purchase Requisition | Heinz Gutschmidt | 02.03.2011 | 20:09:00 | 9 mins | Requester
Filter
Copy
Delete
Export
Version 1.3.4



Disco - Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
219%
search...
Detail
Activities
Paths
608
Create Purchase Requisition
608
374
Analyze Purchase Requisition
382
234
11
8
Amend Purchase Requisition
11
307
3
Create Request for Quotation
544
Leading to End point
Frequency
Show:
Absolute frequency
885
664
442
221
450
337
225
112
Performance
Filter
Animation
Copy
Delete
Export
Version 1.3.4

Results so far...

Original Questions:

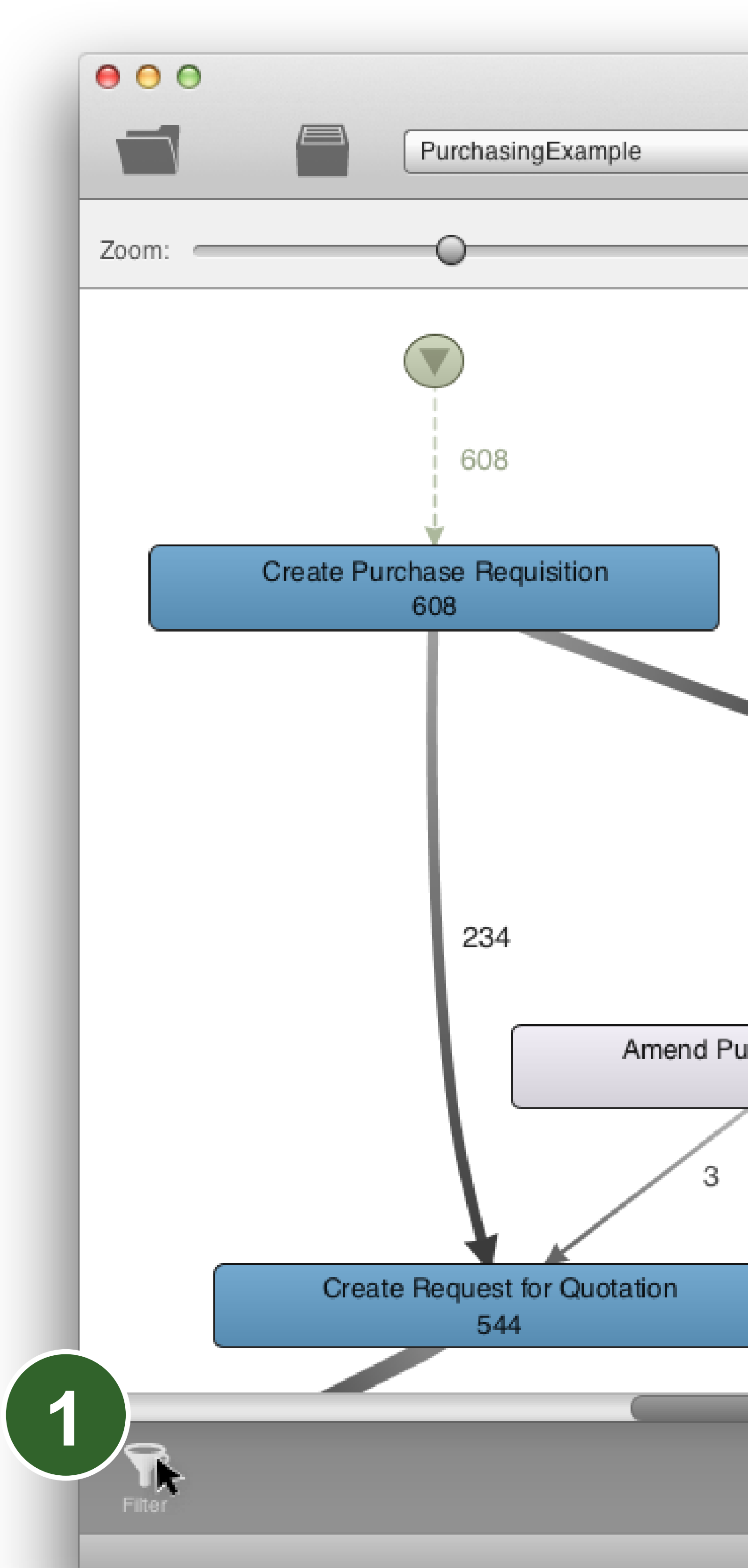
1. How does the process actually look like?
    - Objective process map discovered
    - Lots of amendments and stopped requests: Update of purchasing guidelines needed
2. Are there deviations from the prescribed process?
3. Do we meet the performance targets?
    - Not by all (some take longer than 21 days): Is there a bottleneck in the process? -> **Next**

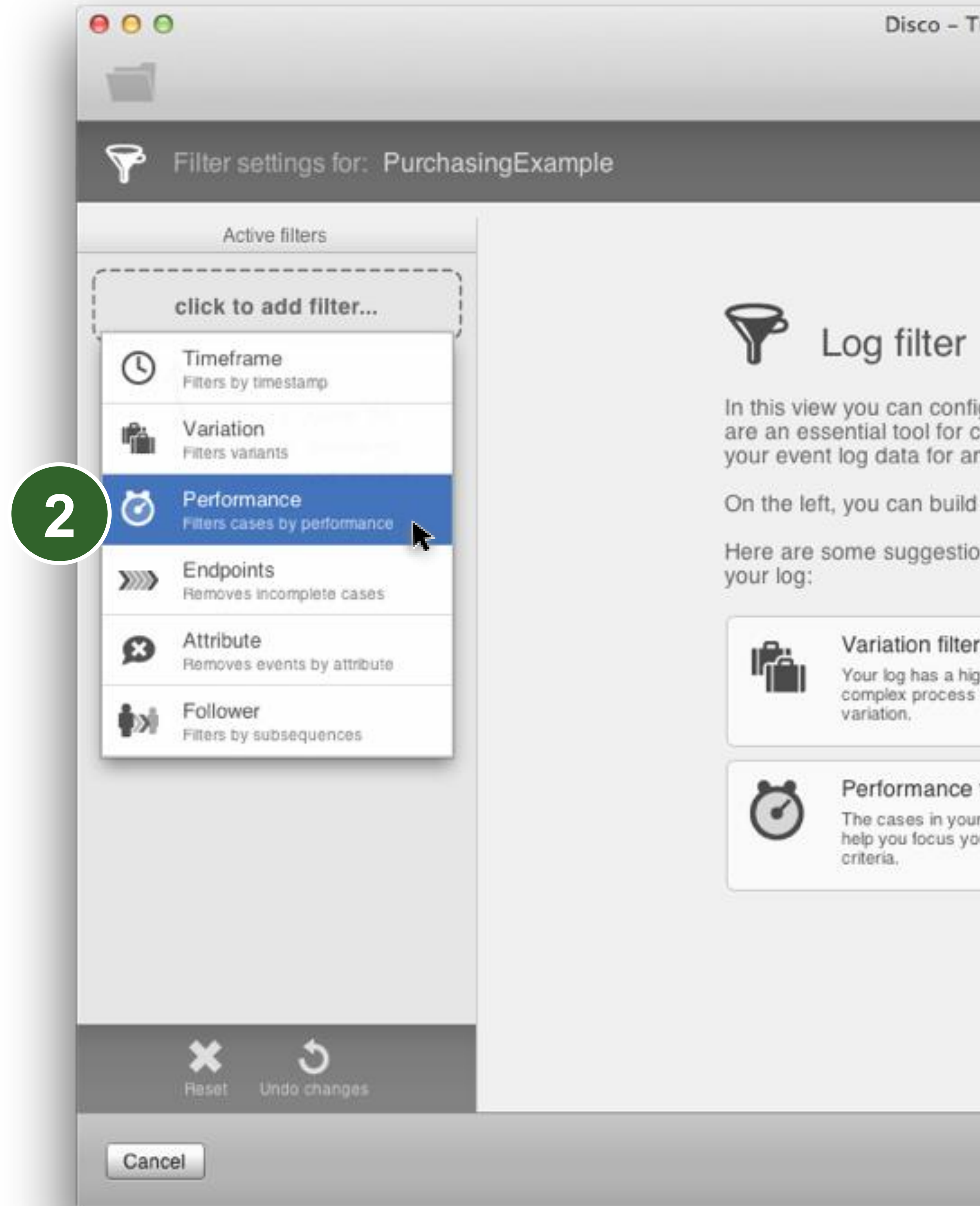
Step 6 - Filter on Performance

Click on the Filter symbol in the lower left corner and add a Performance filter

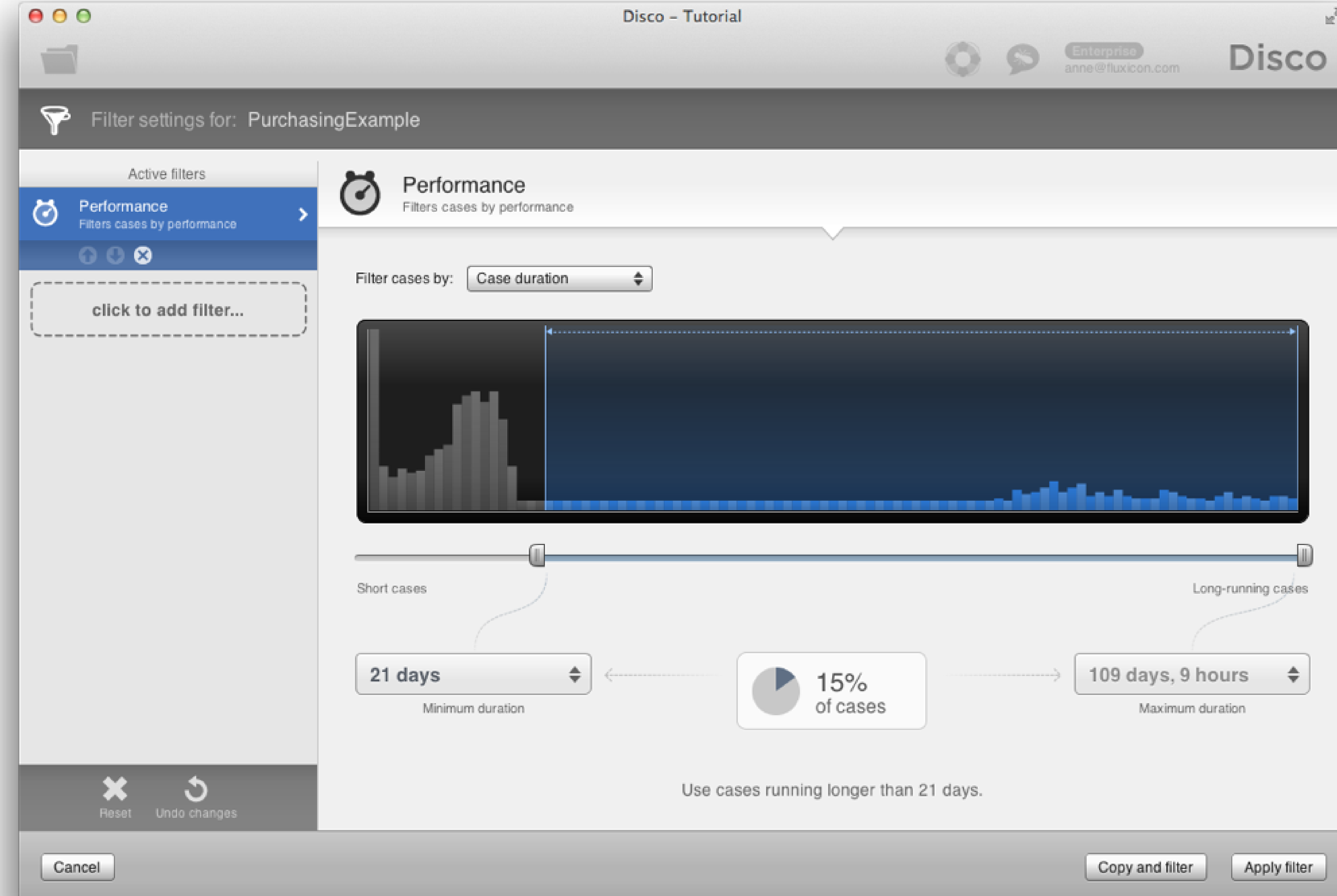
- Select 21 days as lower boundary
- You'll see that ca. 15% of the purchase orders take longer than 21 days

Press 'Apply filter' to focus only on those cases that take longer than 21 days


PurchasingExample
Zoom:
608
Create Purchase Requisition
608
234
Amend Pu
3
Create Request for Quotation
544
1
Filter


Disco – T
Filter settings for: PurchasingExample
Active filters
click to add filter...
Timeframe
Filters by timestamp
Variation
Filters variants
2
Performance
Filters cases by performance
Endpoints
Removes incomplete cases
Attribute
Removes events by attribute
Follower
Filters by subsequences
Reset
Undo changes
Cancel
Log filter
In this view you can confi
are an essential tool for c
your event log data for ar
On the left, you can build
Here are some suggestio
your log:
Variation filter
Your log has a hig
complex process
variation.
Performance
The cases in your
help you focus yo
criteria.

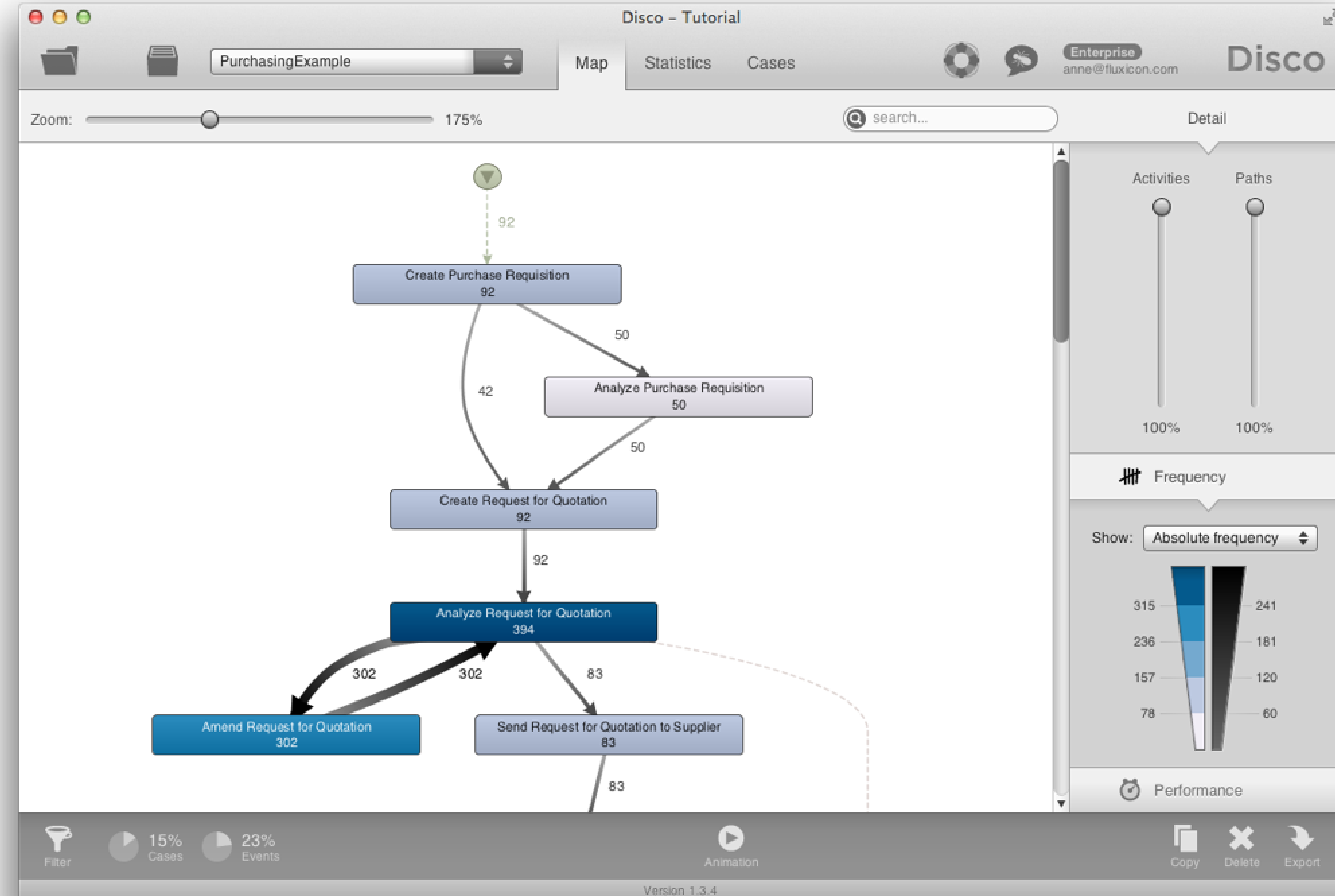
fluxicon


Disco – Tutorial
Enterprise
anne@fluxicon.com
Disco
Filter settings for: PurchasingExample
Active filters
Performance
Filters cases by performance
click to add filter...
Reset
Undo changes
Performance
Filters cases by performance
Filter cases by:
Case duration
Short cases
Long-running cases
21 days
Minimum duration
15%
of cases
109 days, 9 hours
Maximum duration
Use cases running longer than 21 days.
Cancel
Copy and filter
Apply filter

Step 7 - Spot Bottlenecks

The filtered process map shows the process flow for the 92 (15%) ‘slow’ cases

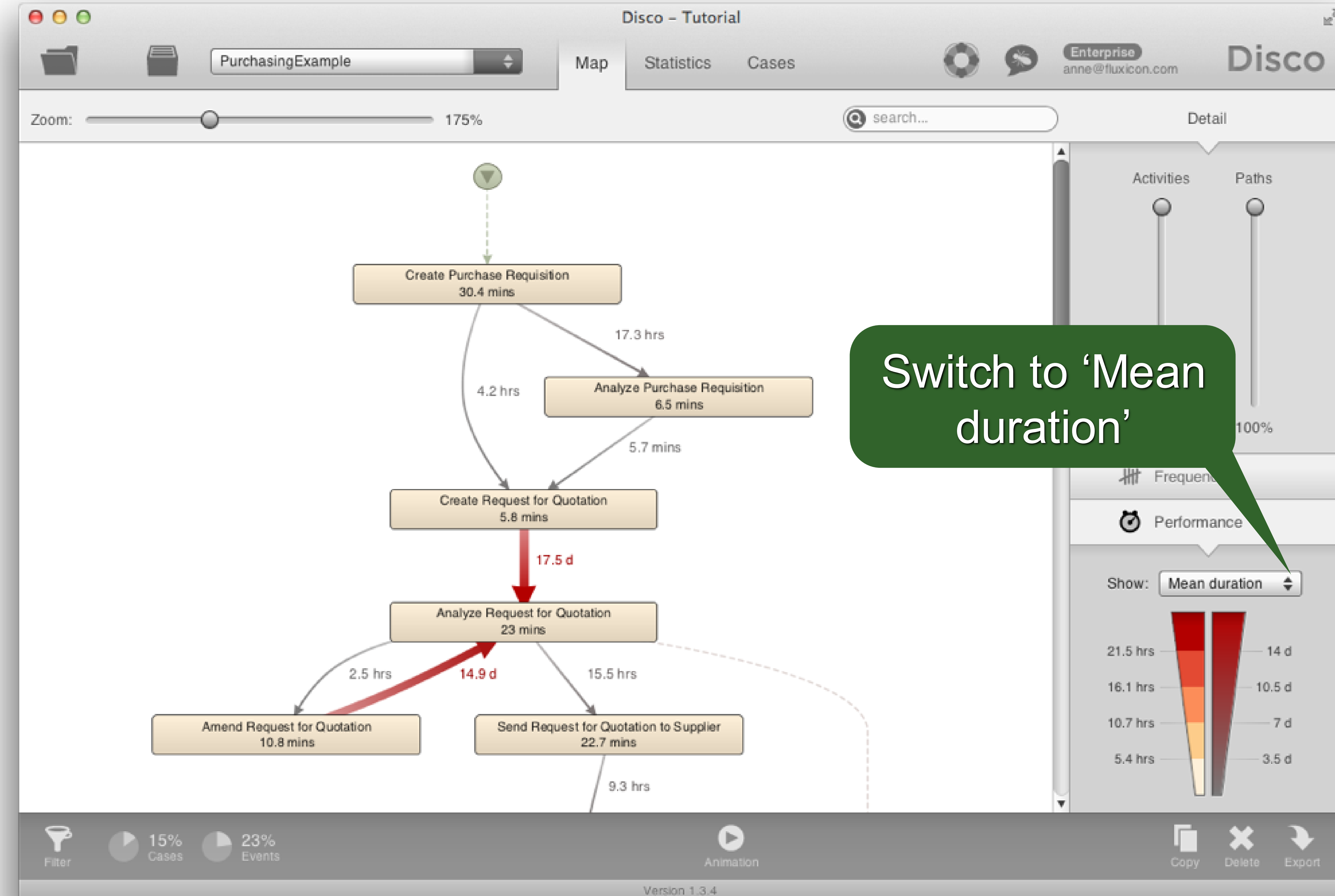
- On average 3 amendments per case!
  - 92 cases, 302 amendments…


Disco – Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
175%
search...
Detail
92
Create Purchase Requisition
92
50
42
Analyze Purchase Requisition
50
50
Create Request for Quotation
92
92
Analyze Request for Quotation
394
302
302
83
Amend Request for Quotation
302
Send Request for Quotation to Supplier
83
83
Activities
Paths
100%
100%
Frequency
Show:
Absolute frequency
315
236
157
78
241
181
120
60
Performance
Filter
15%
Cases
23%
Events
Animation
Copy
Delete
Export
Version 1.3.4

Step 7b - Spot Bottlenecks

Switch to ‘Performance’ view

- ‘Total duration’ shows the high-impact areas
- Switch to ‘Mean duration’:
    - On average it takes **more than 14 days** to return from the rework loop to the normal process
    - What about min and max duration?


Disco - Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
175%
search...
Detail
Activities
Paths
Create Purchase Requisition
30.4 mins
17.3 hrs
4.2 hrs
Analyze Purchase Requisition
6.5 mins
5.7 mins
Create Request for Quotation
5.8 mins
17.5 d
Analyze Request for Quotation
23 mins
2.5 hrs
14.9 d
15.5 hrs
Amend Request for Quotation
10.8 mins
Send Request for Quotation to Supplier
22.7 mins
9.3 hrs
100%
Performance
Show:
Mean duration
21.5 hrs
16.1 hrs
10.7 hrs
5.4 hrs
14 d
10.5 d
7 d
3.5 d
Filter
15%
Cases
23%
Events
Animation
Copy
Delete
Export
Version 1.3.4
Switch to 'Mean duration'

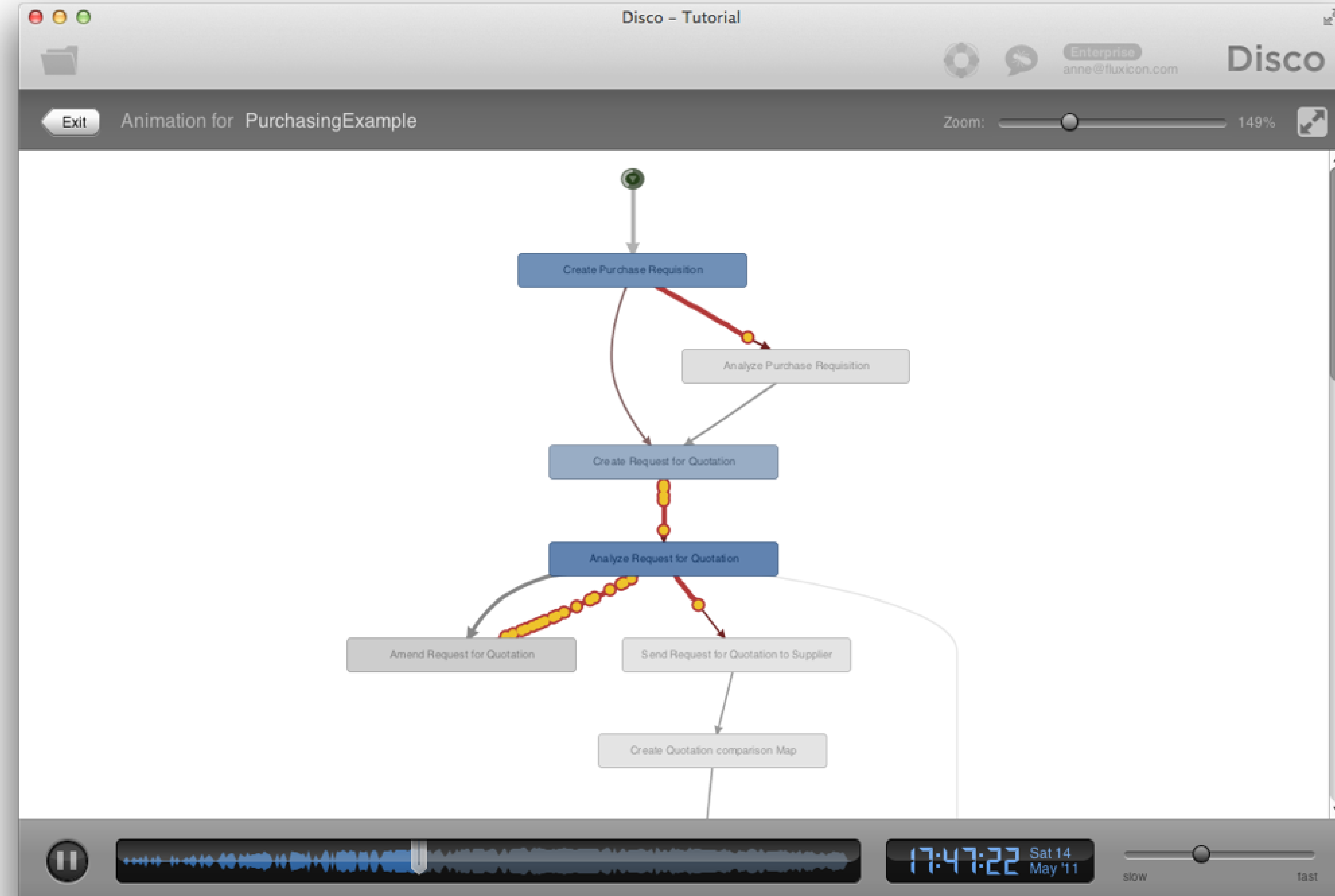
Step 8 - Animate Process

Visualize bottleneck:
Press ▷ button to start animation

Observe how purchase orders move through the process

Drag needle to the end of the timeline

- observe how the most used paths get thicker and thicker



Disco – Tutorial
Enterprise
anne@fluxicon.com
Disco
Exit
Animation for PurchasingExample
Zoom:
149%
Create Purchase Requisition
Analyze Purchase Requisition
Create Request for Quotation
Analyze Request for Quotation
Amend Request for Quotation
Send Request for Quotation to Supplier
Create Quotation comparison Map
17:47:22
Sat 14
May '11
slow
fast

Results so far...

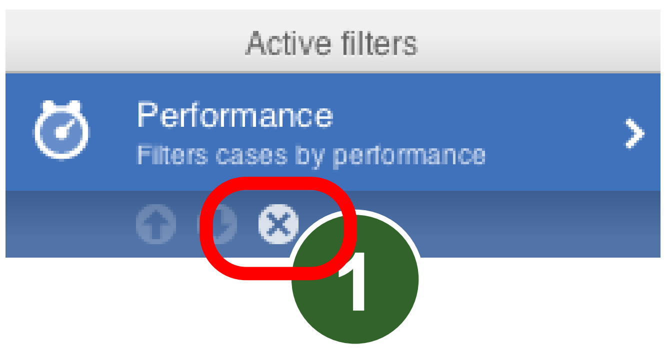
1. How does the process actually look like?
   - Objective process map discovered
   - Lots of amendments and stopped requests: Update of purchasing guidelines needed

2. Are there deviations from the prescribed process? -> **Next**

3. Do we meet the performance targets?
   - Not by all (some take longer than 21 days)
   - The ‘Analyze Request for Quotation’ activity is a huge bottleneck: Process change is needed

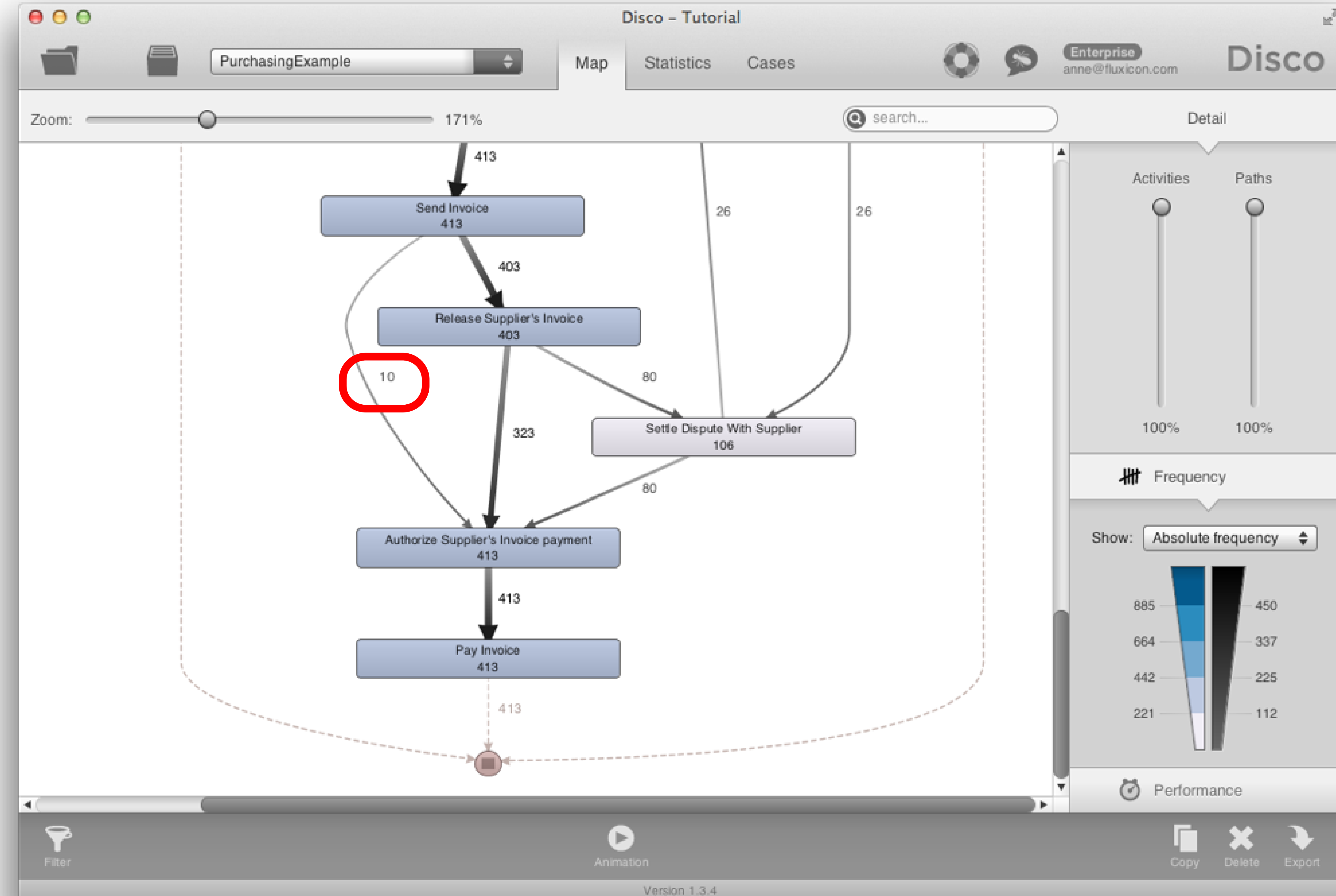
Step 9 - Compliance Check

Exit the animation, return to Filter settings, and remove performance filter



Switch back to Frequency Map view and scroll to end of the process

- 10 cases skip the mandatory ‘Release Supplier’s Invoice’ activity!



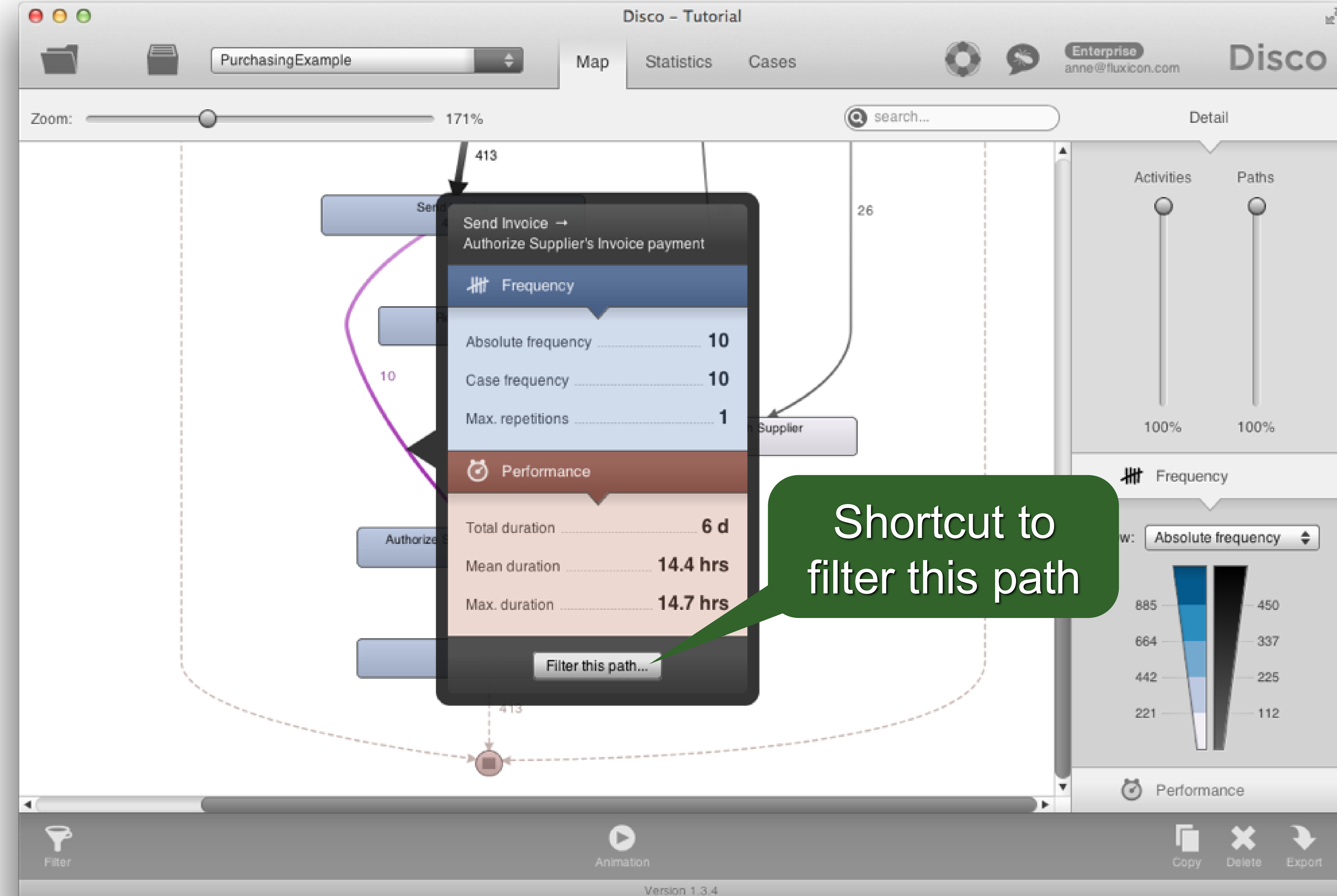
Disco – Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
171%
search...
Detail
413
Send Invoice
413
403
Release Supplier's Invoice
403
10
80
323
Settle Dispute With Supplier
106
26
26
80
Authorize Supplier's Invoice payment
413
413
Pay Invoice
413
413
Activities
Paths
100%
100%
Frequency
Show:
Absolute frequency
885
664
442
221
450
337
225
112
Performance
Filter
Animation
Copy
Delete
Export
Version 1.3.4

Step 9 - Compliance Check

Drill down: Click on the path from ‘Send invoice’ to ‘Authorize Supplier’s Invoice payment’ and press ‘Filter this path...’

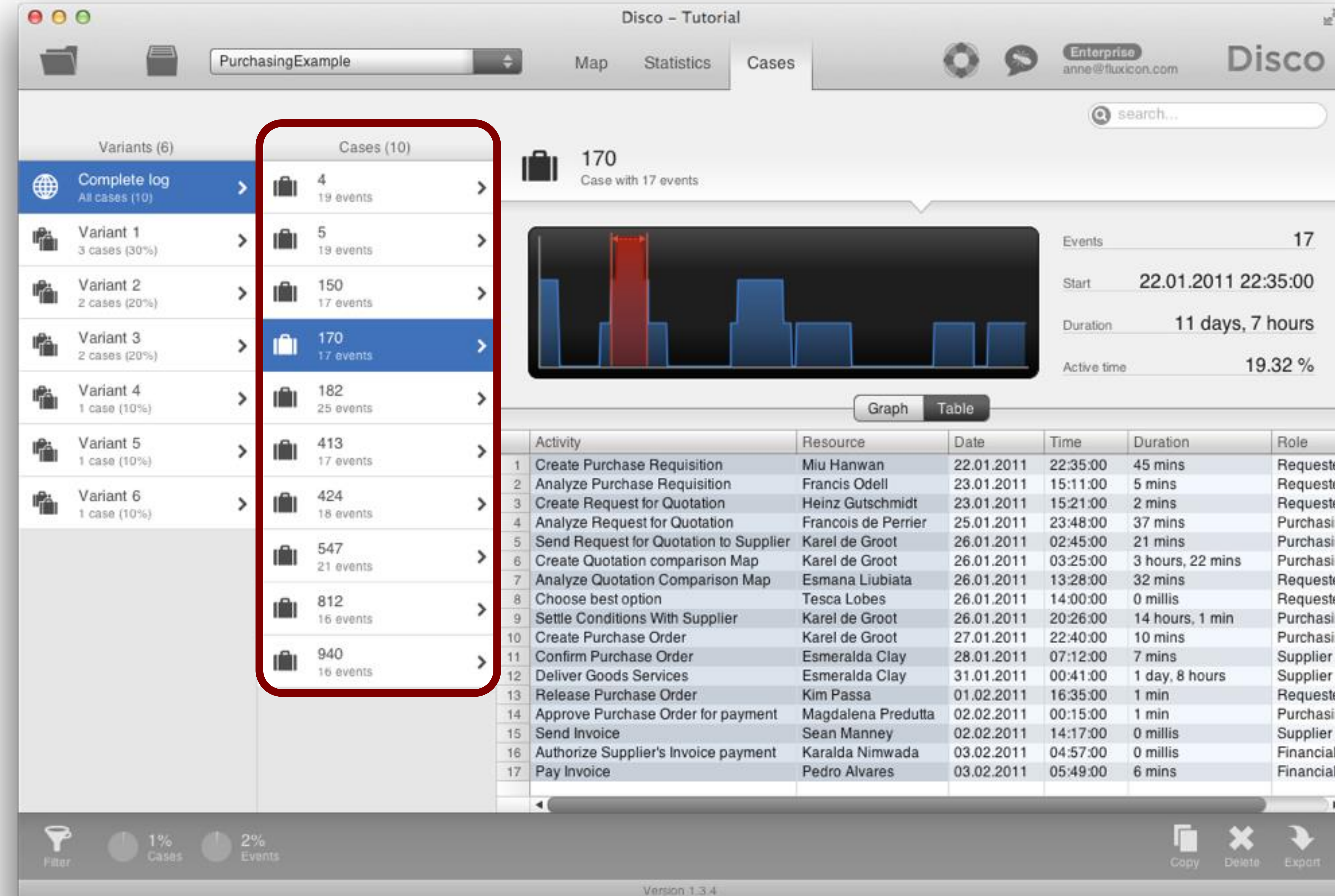
Switch to Cases view to see the 10 cases

- Actionable result: We can either change the operational system to prevent the violation or provide targeted training



Disco - Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
171%
search...
Detail
413
26
Send Invoice →
Authorize Supplier's Invoice payment
Frequency
Absolute frequency 10
Case frequency 10
Max. repetitions 1
Performance
Total duration 6 d
Mean duration 14.4 hrs
Max. duration 14.7 hrs
Filter this path...
10
Activities
Paths
100%
100%
Frequency
Absolute frequency
885
664
442
221
450
337
225
112
Performance
Filter
Animation
Copy
Delete
Export
Version 1.3.4
Shortcut to filter this path

fluxicon



Disco – Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
search...
Variants (6)
Complete log
All cases (10)
Variant 1
3 cases (30%)
Variant 2
2 cases (20%)
Variant 3
2 cases (20%)
Variant 4
1 case (10%)
Variant 5
1 case (10%)
Variant 6
1 case (10%)
Cases (10)
4
19 events
5
19 events
150
17 events
170
17 events
182
25 events
413
17 events
424
18 events
547
21 events
812
16 events
940
16 events
170
Case with 17 events
Events 17
Start 22.01.2011 22:35:00
Duration 11 days, 7 hours
Active time 19.32 %
Graph
Table
Activity
Resource
Date
Time
Duration
Role
1 Create Purchase Requisition Miu Hanwan 22.01.2011 22:35:00 45 mins Requester
2 Analyze Purchase Requisition Francis Odell 23.01.2011 15:11:00 5 mins Requester
3 Create Request for Quotation Heinz Gutschmidt 23.01.2011 15:21:00 2 mins Requester
4 Analyze Request for Quotation Francois de Perrier 25.01.2011 23:48:00 37 mins Purchasing
5 Send Request for Quotation to Supplier Karel de Groot 26.01.2011 02:45:00 21 mins Purchasing
6 Create Quotation comparison Map Karel de Groot 26.01.2011 03:25:00 3 hours, 22 mins Purchasing
7 Analyze Quotation Comparison Map Esmana Liubiata 26.01.2011 13:28:00 32 mins Requester
8 Choose best option Tesca Lobes 26.01.2011 14:00:00 0 millis Requester
9 Settle Conditions With Supplier Karel de Groot 26.01.2011 20:26:00 14 hours, 1 min Purchasing
10 Create Purchase Order Karel de Groot 27.01.2011 22:40:00 10 mins Purchasing
11 Confirm Purchase Order Esmeralda Clay 28.01.2011 07:12:00 7 mins Supplier
12 Deliver Goods Services Esmeralda Clay 31.01.2011 00:41:00 1 day, 8 hours Supplier
13 Release Purchase Order Kim Passa 01.02.2011 16:35:00 1 min Requester
14 Approve Purchase Order for payment Magdalena Predutta 02.02.2011 00:15:00 1 min Purchasing
15 Send Invoice Sean Manney 02.02.2011 14:17:00 0 millis Supplier
16 Authorize Supplier's Invoice payment Karalda Nimwada 03.02.2011 04:57:00 0 millis Financial
17 Pay Invoice Pedro Alvares 03.02.2011 05:49:00 6 mins Financial
Filter
1% Cases
2% Events
Copy
Delete
Export
Version 1.3.4

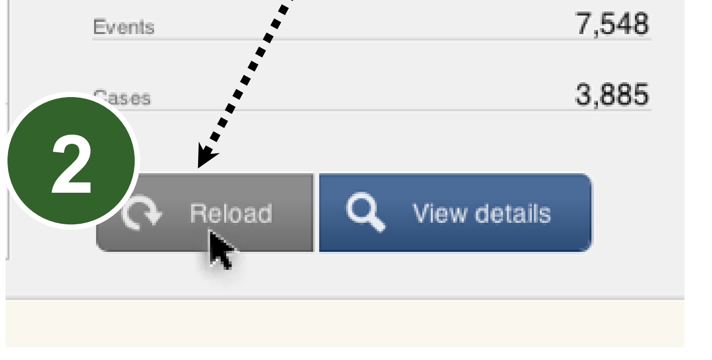
Results so far...

1. How does the process actually look like?
    - Objective process map discovered
    - Lots of amendments and stopped requests: Update of purchasing guidelines needed
2. Are there deviations from the prescribed process? -> Yes, training or system change needed
3. Do we meet the performance targets?
    - Not by all (some take longer than 21 days)
    - The ‘Analyze Request for Quotation’ activity is a huge bottleneck: Process change is needed

Step 10 - Organizational View

Last Step: We seek an alternative view on the data to visualize the organizational flow

Go to ‘Project view’ and press ‘Reload’:



Set ‘Activity’ column to ‘Other’ and configure ‘Role’ column as ‘Activity’

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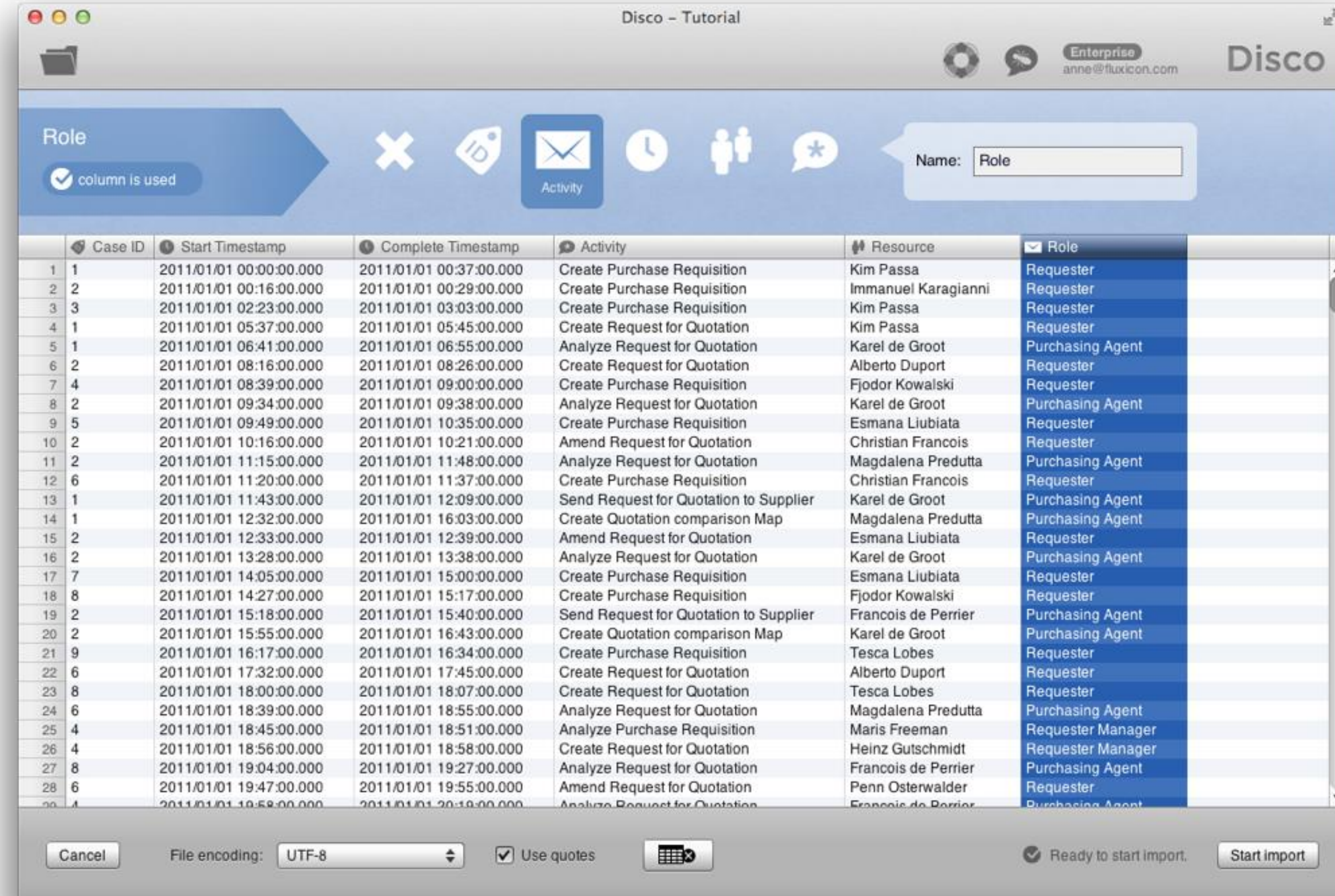

	Case ID	Start Timestamp	Complete Timestamp	Activity	Resource	Role
1	1	2011/01/01 00:00:00.000	2011/01/01 00:37:00.000	Create Purchase Requisition	Kim Passa	Requester
2	2	2011/01/01 00:16:00.000	2011/01/01 00:29:00.000	Create Purchase Requisition	Immanuel Karagianni	Requester
3	3	2011/01/01 02:23:00.000	2011/01/01 03:03:00.000	Create Purchase Requisition	Kim Passa	Requester
4	1	2011/01/01 05:37:00.000	2011/01/01 05:45:00.000	Create Request for Quotation	Kim Passa	Requester
5	1	2011/01/01 06:41:00.000	2011/01/01 06:55:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
6	2	2011/01/01 08:16:00.000	2011/01/01 08:26:00.000	Create Request for Quotation	Alberto Duport	Requester
7	4	2011/01/01 08:39:00.000	2011/01/01 09:00:00.000	Create Purchase Requisition	Fjodor Kowalski	Requester
8	2	2011/01/01 09:34:00.000	2011/01/01 09:38:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
9	5	2011/01/01 09:49:00.000	2011/01/01 10:35:00.000	Create Purchase Requisition	Esmana Liubiata	Requester
10	2	2011/01/01 10:16:00.000	2011/01/01 10:21:00.000	Amend Request for Quotation	Christian Francois	Requester
11	2	2011/01/01 11:15:00.000	2011/01/01 11:48:00.000	Analyze Request for Quotation	Magdalena Predutta	Purchasing Agent
12	6	2011/01/01 11:20:00.000	2011/01/01 11:37:00.000	Create Purchase Requisition	Christian Francois	Requester
13	1	2011/01/01 11:43:00.000	2011/01/01 12:09:00.000	Send Request for Quotation to Supplier	Karel de Groot	Purchasing Agent
14	1	2011/01/01 12:32:00.000	2011/01/01 16:03:00.000	Create Quotation comparison Map	Magdalena Predutta	Purchasing Agent
15	2	2011/01/01 12:33:00.000	2011/01/01 12:39:00.000	Amend Request for Quotation	Esmana Liubiata	Requester
16	2	2011/01/01 13:28:00.000	2011/01/01 13:38:00.000	Analyze Request for Quotation	Karel de Groot	Purchasing Agent
17	7	2011/01/01 14:05:00.000	2011/01/01 15:00:00.000	Create Purchase Requisition	Esmana Liubiata	Requester
18	8	2011/01/01 14:27:00.000	2011/01/01 15:17:00.000	Create Purchase Requisition	Fjodor Kowalski	Requester
19	2	2011/01/01 15:18:00.000	2011/01/01 15:40:00.000	Send Request for Quotation to Supplier	Francois de Perrier	Purchasing Agent
20	2	2011/01/01 15:55:00.000	2011/01/01 16:43:00.000	Create Quotation comparison Map	Karel de Groot	Purchasing Agent
21	9	2011/01/01 16:17:00.000	2011/01/01 16:34:00.000	Create Purchase Requisition	Tesca Lobes	Requester
22	6	2011/01/01 17:32:00.000	2011/01/01 17:45:00.000	Create Request for Quotation	Alberto Duport	Requester
23	8	2011/01/01 18:00:00.000	2011/01/01 18:07:00.000	Create Request for Quotation	Tesca Lobes	Requester
24	6	2011/01/01 18:39:00.000	2011/01/01 18:55:00.000	Analyze Request for Quotation	Magdalena Predutta	Purchasing Agent
25	4	2011/01/01 18:45:00.000	2011/01/01 18:51:00.000	Analyze Purchase Requisition	Maris Freeman	Requester Manager
26	4	2011/01/01 18:56:00.000	2011/01/01 18:58:00.000	Create Request for Quotation	Heinz Gutschmidt	Requester Manager
27	8	2011/01/01 19:04:00.000	2011/01/01 19:27:00.000	Analyze Request for Quotation	Francois de Perrier	Purchasing Agent
28	6	2011/01/01 19:47:00.000	2011/01/01 19:55:00.000	Amend Request for Quotation	Penn Osterwalder	Requester

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Step 10 - Organizational View

Instead of the activity flow, we are now looking at how the process moves through different *roles* in the organization

- Inefficiencies can often be found at the borders of organizational units
- Clearly, the Purchasing agents are causing the biggest delays in the process!

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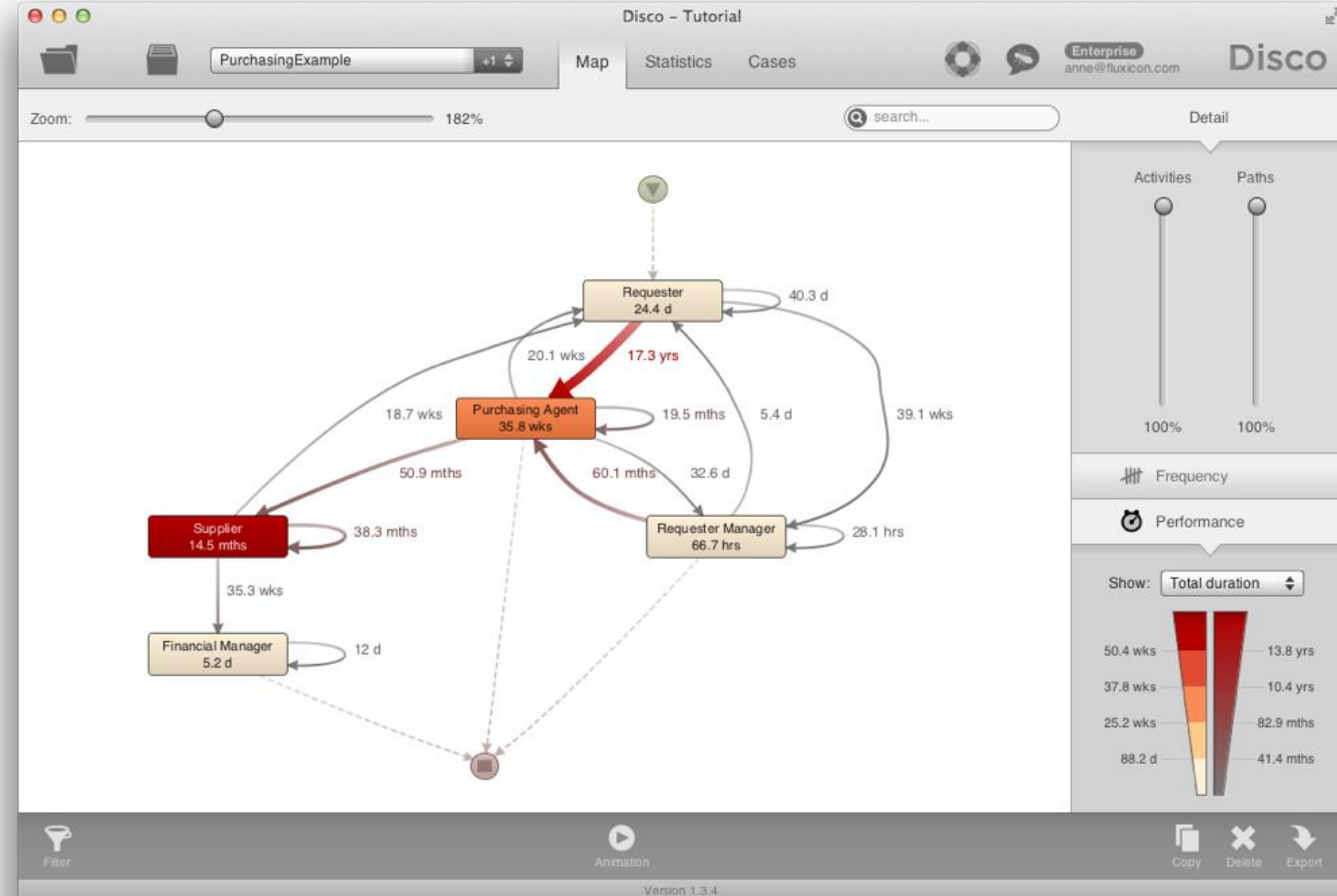


Disco - Tutorial
PurchasingExample
Map
Statistics
Cases
Enterprise
anne@fluxicon.com
Disco
Zoom:
182%
search...
Detail
Activities
Paths
100%
100%
Frequency
Performance
Show:
Total duration
50.4 wks
37.8 wks
25.2 wks
88.2 d
13.8 yrs
10.4 yrs
82.9 mths
41.4 mths
Requester
24.4 d
40.3 d
20.1 wks
17.3 yrs
18.7 wks
Purchasing Agent
35.8 wks
19.5 mths
5.4 d
39.1 wks
50.9 mths
60.1 mths
32.6 d
Supplier
14.5 mths
38.3 mths
Requester Manager
66.7 hrs
28.1 hrs
35.3 wks
Financial Manager
5.2 d
12 d
Filter
Animation
Copy
Delete
Export
Version 1.3.4

Close the loop

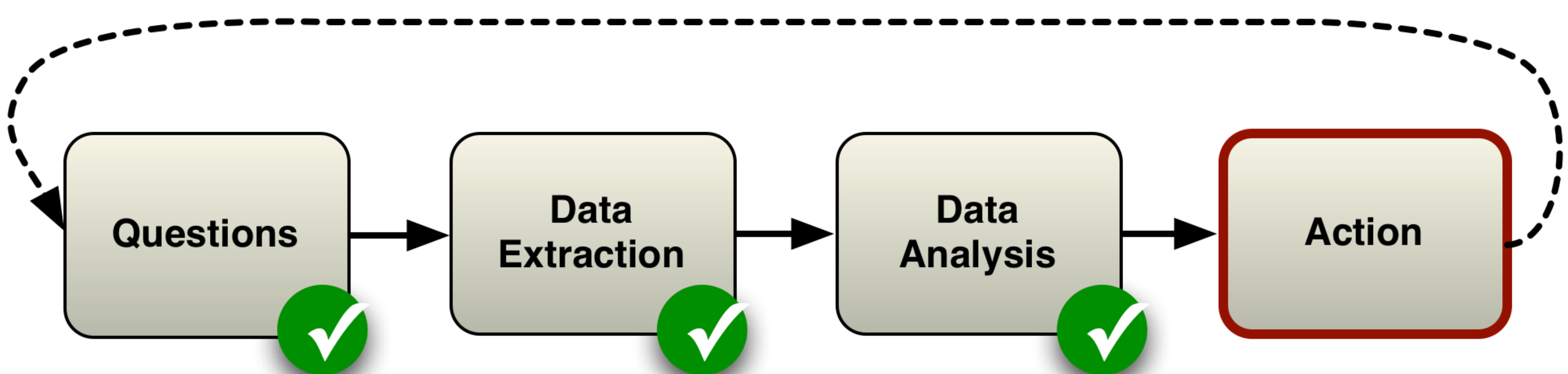
We have learned

- How to play with Disco to discover a process model

We have found

- opportunities to improve the process

Now: Take action and verify results



Bonus

Import data again and configure both 'Activity' and 'Role' column as 'Activity'

- Can you see what happens now?

Outline

0. What is Process mining?
1. Example Scenario
2. Roadmap
3. Hands-on Session
4. **Take-away Points**

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Take-away Points

Real processes are often more complex than you would expect

There is no one “right” model:

- You can take multiple views on the same data

Process mining is an explorative, interactive activity

Questions?

Want to use these techniques in your research?

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